

Research Report on

Mechanistic evidence and causal tests for AI intentional semantics

Full search query: I want to find work that assesses whether contemporary AI systems—especially large language models and their multi-modal/agentic variants—possess genuine semantics (intentionality) by examining under the hood mechanistic evidence (internal representations, circuits, and causal mediation of reference and truth-conditional content), alongside proposed tests, grounding criteria, and counterarguments

Summary

The literature provides strong, selective, and path localized mechanistic evidence that contemporary LLMs implement internal variables that causally mediate reference (binding, entity tracking, belief visibility) via identifiable subspaces and circuits [1,4,6,14], while evidence that they similarly mediate truth conditional content is growing but less complete—sufficiency is robust via directional/cone steering [2,3,17,18] and partial mediation under context [7,8], yet full path level necessity and circuit decomposition remain comparatively sparse; multimodal work extends causal mediation to cross modal reference via spatial/grounding IDs and cross attention heads [10,11,12,13,12], but has not yet established strong truth conditional mediation.

What constitutes the strongest mechanistic evidence so far

- Causally necessary-and-sufficient internal variables for reference:
 - Binding/Ordering IDs in residual stream low rank subspaces that index entities/attributes and mediate retrieval, validated with activation patching and selective interchange interventions [4,14,1,6].
 - Causal abstraction alignment of high level belief/reference models to internal activations, with variable level and hidden feature interchanges that flip predicted beliefs while preserving unrelated behavior [5,1].
 - Decomposition into distinct, layer localized mechanisms for retrieval (positional, lexical, reflexive) with a mixture model that predicts logits and targeted ablations/interchanges confirming necessity and path specificity [6].
 - Belief visibility and lookback circuits encoding who saw what and binding answer spans, with Ordering IDs co locating reference information in a low rank subspace; validated by swapping identical surface forms with different IDs [1].

Truth-conditional content: what is established and what is missing

- Evidence for sufficiency and structure:
 - Truth “directions” and, more broadly, multidimensional “truth cones” whose activation steering or directional ablations reliably flip true/false judgments while preserving other capabilities; generalizes across models to varying degrees [2,3,17,18].
 - Context sensitive belief/truth directions that mediate in context inference to a measurable extent; interventions moving premises along these directions shift related hypotheses accordingly, indicating partial causal mediation [7,8].
 - Earlier dynamic semantics studies show that proposition truth over entities is decodable and edits have predictable generation effects [19]; compositional “propositional probes” bind predicates and arguments via a binding subspace and can monitor latent world states diverging from outputs [20].
- Limitations and open issues:
 - Necessity and path specific mediation for truth conditions are weaker than for reference: most truth results show sufficiency via steering rather than identifying minimal, necessary circuits or path restricted causal chains [2,3,17,18].
 - Confounds between “truth” and common belief/verifiability remain in some datasets; authors note scope limits to simple factual sentences [3].
 - Internal–external misalignment: models may internally encode correct answers while generating incorrect ones, underscoring a separation between content representations and response policy [23].

Multimodal grounding: causal mediation of reference, not yet of propositions

- Established mechanisms:
 - Binding/grounding IDs that link image tokens to textual referents, with patch level concept localization and causal interventions confirming mediation [10,13].
 - Position ID heads and feature retrieval heads implementing spatial indexing; targeted head level interventions induce systematic binding errors, indicating necessity and path specificity [11].
 - Cross attention heads in middle layers of BLIP/LLaVA causally mediate object grounding under semantically controlled image/text corruptions (NOTICE), avoiding noise corruption artifacts [12]; visual tokens feed language vocabulary alignment with strong token ablation effects [28].
- Gaps:
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Current multimodal results focus on referential specificity and constraint satisfaction; circuit level mediation of truth conditional propositions across modalities is still largely untested [10,11,12,13,12,28,30].

Mechanistic reasoning circuits relevant to semantic composition

- Necessary and sufficient circuits have been isolated for symbolic reasoning tasks:
 - Propositional microtasks and syllogistic inference show sparse, decomposable circuits validated by activation patching and targeted ablations, including stage wise processing and query triggered chains; some transfer to pretrained LLMs [25,27].
 - Synthetic “world model” settings (e.g., Othello) demonstrate that counterfactual activation edits can causally control next move probabilities, although linear decodability may be weak and circuit localization incomplete [26].
- Relevance: These studies establish that LMs can implement minimal circuits for content bearing variables and logical composition, suggesting a methodology for tracing truth conditional mediation in more naturalistic settings.

Methodological advances, controls, and pitfalls

- Frameworks that strengthen causal claims:
 - Causal abstraction with interchange interventions to validate that aligned internal variables have the same causal properties as high level semantic variables [5], instantiated at scale for belief/reference [1,6].
 - Latent causal probing formalizes probe validity within SCMs and uses causal mediation analysis to isolate the model's causal path, reducing probe induced confounds in grounded synthetic domains [9].
- Validity concerns and recommended controls:
 - High dimensional interventional probing can fail due to redundancy and rank confounds; strong baselines and counterfactual vector interchange controls are needed [32].
 - Probe only decodability is insufficient; selective necessity/sufficiency via activation/weight/path patching, interchange interventions, and causal scrubbing should be prioritized [5,1,6,27].
 - Distinguish content from policy layers and control for instruction tuning artifacts; misalignment work shows internal truth signals can be suppressed behaviorally [23,20].

Overall assessment for intentionality

- Reference/intentionality: The most compelling evidence for genuine semantics (operationalized as internal, causally mediating referential content) comes from binding/ID subspaces and belief tracking circuits with selective, path localized causal tests and counterfactual interchanges [1,4,6,14].
- Truth conditions: There is converging evidence that models encode truth conditional polarity in structured subspaces (directions/cones) that can be causally steered, and that these signals participate in in context inference [2,3,7,8,17,18]. However, compared with reference, fewer works decompose the full causal chain for propositional truth with strong necessity and path specificity; bridging this gap is an important next step.
- Multimodal grounding: Clear causal mediation of cross modal reference has been demonstrated, but truth conditional mediation across modalities remains open [10,11,12,13,12].

High value directions for follow up experiments

- Unify binding IDs with propositional truth: use closed world microworlds with ground truth latent states to perform interchange interventions that swap specific predicate truth values while holding referents and surfaces constant; verify necessity/sufficiency and path specificity with causal scrubbing and minimal path patching [5,1,20,25,27].
- Move from directions to circuits for truth: start from truth cones [2] to discover sparse pathways via path restricted patching and head/MLP level ablations; test robustness under negation, modality, and quantifier scope with counterfactual minimal pairs [3,7,8,27].
- Cross modal truth mediation: adapt NOTICE style semantically controlled corruptions to proposition level VQA (e.g., counterfactual image edits that flip predicate truth) and validate mediation via cross attention head interventions plus cross modal interchange (swap visual token subspaces corresponding to the proposition) [12,10,11,12,13].
- Disentangle content vs policy: combine propositional probes with response policy disentanglement and targeted steering to align internal truth states to outputs without degrading unrelated behavior; measure selectivity under paraphrase and adversarial distractors [20,23,17].

Categories

Comparative overview of mechanistic evidence for semantics/intentionality in contemporary AI

The tables below compare core aspects an expert would care about: what semantic variables are targeted; whether internal features/circuits are localized; the strength and selectivity of causal mediation evidence (necessity, sufficiency, path specificity); grounding routes (text, multimodal, interactive); robustness and controls; and key limitations. Citations are provided inline.

Legend for causal evidence

- Necessity: removing/ablating a feature/circuit degrades the targeted semantic behavior.
- Sufficiency: forcing/steering a feature/circuit induces the targeted semantic behavior in its absence.
- Path specificity: mediation localized to hypothesized pathways (e.g., layer/heads, restricted patching).
- Selectivity: intervention preserves unrelated behavior while changing the targeted semantic variable.

High-level comparison table: scope, interventions, and mediation strength

Paper	Semantic target	Modality	Main mechanistic methods	Causality strength	Data scope	Robustness/controls	Key limitation(s)
[1]	Belief tracking; entity/object/state reference via Ordering IDs (OIs); visibility relation	Text	Causal mediation tracing; causal abstraction alignment; interchange interventions at high-level vars and hidden features	Necessity, sufficiency, path specificity, selectivity (IIA metric)	Synthetic ToM microstories; Llama-3/3.1; generalization to BigToM	Counterfactual minimal pairs; swap identical surface forms with different OIs	Focused on synthetic ToM; external validity to open-domain untested
[2]	Truthfulness as multidimensional “truth cones”	Text	Activation steering; directional ablation along cone bases	Sufficiency; selectivity (preserve unrelated behavior)	Multiple LLM families; simple factual statements	Cross-architecture generalization claims; sparsity/disentanglement analysis	Truth scope limited to simple propositions; axes’ semantics partially opaque
[3]	Linear “truth direction”	Text	Probing (mass-mean); localized activation interventions near final tokens	Sufficiency; some localization	LLaMA variants; curated true/false datasets	Probe transfer across datasets; note confounds (common belief vs truth)	Limited model coverage; truth conflated with verifiability/common belief; necessity not fully shown
[4]	Binding IDs for entity–attribute binding	Text	Activation patching/interventions; subspace identification	Necessity (binding IDs affect behavior)	Pythia/LLaMA; in-context binding tasks	Distance in subspace reflects binding discriminability	Circuit generation/use left for future work; larger models only
[6]	Retrieval mechanisms (positional, lexical, reflexive) for bound entities	Text	Interchange interventions; layer-wise patching/ablation; causal abstraction mixture model	Necessity (layer-specific), path specificity; high agreement (~95%)	9 models; 10 binding tasks; long contexts	Counterfactual pairs distinguishing mechanisms	Scope is retrieval, not broader truth/proposition semantics
[7,8]	Belief/truth-value directions under context	Text	Probing (logistic; CCS/MMP); intervention moving premise along belief direction and measuring hypothesis shift	Partial mediation (“one of” causal mediators); context sensitivity analysis	7B/13B; pretrained/instruction-tuned	Layer/model/data dependence; error types and responsiveness quantified	Deeper mechanistic pathways not localized; work-in-progress
[9]	Latent causal probing of SCM variables (grid-world semantics)	Text	Probing framed via SCM; causal mediation analysis excluding probe influence	Necessity/mediation under SCM assumptions	Synthetic grid-world with ground-truth states	Formal control guarantees; generalization to novel sequences	Synthetic domain; scope limited beyond LMs’ open-world semantics
[10]	Visual–text binding via Binding IDs	Multimodal	Adapted causal mediation (activation patching) on VLM residual streams	Necessity (concept tokens carry info; binding vectors mediate)	Shapes QA synthetic; standard VLM pipeline	Multiscale tokens; concept localization	Focused on binding; truth/propositional content not analyzed
[11]		Multimodal					

	Position ID heads; feature retrieval heads for spatial binding		Causal mediation analysis on attention-head outputs; targeted head interventions	Necessity (binding errors via interventions); path specificity	VLMs (not specified beyond class)	Interference sources identified	No analysis of truth-conditional content; content-independent binding focus
[12]	Cross-attention heads mediating object detection/suppression (image grounding)	Multimodal	Activation patching with semantically controlled corruptions (SIP/STR)	Path specificity (middle-layer cross-attention); necessity via patching effects	BLIP, LLaVA; curated SIP/STR datasets	Avoids Gaussian noise pitfalls; code released	Functional categorization; not propositional truth/reference tests
[13]	Grounding IDs induced by external cues (reduce modality gap; mediate binding)	Multimodal	Attention analysis; layerwise probes; ablations; causal interventions	Necessity (binding improvement mediated by IDs)	Synthetic partitioned images; Qwen2.5-VL	Visual+text cue ablation; hallucination reduction claims	Focused on binding; lacks truth-conditional evaluation
[14,16]	Ordering IDs (OI) subspace as BI prototype; causal edits	Text	PCA/ICA subspace discovery; activation editing/patching	Sufficiency (change binding by editing PCs)	Synthetic entity tracking; multiple LMs	Code/datasets; layer focus on query	No circuit identification; synthetic scope; indirect BI-OI interaction
[15]	Directions for self/other beliefs in attention-head activation space	Text	Linear probes on head activations; vector-space interventions	Sufficiency (large behavioral changes); some generalization across ToM tasks	Mistral-7B-Instruct	Gradient mapping to causal tokens; counterfactual manipulations	Single family; task types limited
[17]	Truthful vs hallucination direction (autoencoder-derived)	Text	Representation split; contrastive learning; inference-time vector editing	Sufficiency; selectivity (preserve generation ability)	13 LLMs; TruthfulQA	Opposite-direction edits induce hallucination; transfer across models	Direction semantics opaque; not localized circuits
[18]	Adaptive activation steering for truthfulness	Text	Head-level probes; additive steering; adaptive intensity	Sufficiency; broad applicability	Many LLaMA-family models	Diverse hallucination categories	Probes may capture correlated signals; necessity not shown
[19]	Dynamic semantics: proposition truth over entities	Text	Linear probes; representation editing/manipulation	Sufficiency (predictable generation changes); localization	Alchemy, TextWorld, synthetic	Argue pretraining drives phenomena	Primarily probing; limited causal path tests
[20]	Propositional probes: predicate-argument binding	Text	Compositional decoding via binding subspace	Decoding; monitoring unfaithfulness	Templated contexts; stories; Spanish	Closed-world validation; generalization	Interventions not detailed in excerpts
[21]	ToM belief representations; brittleness; activation edits	Text	Residual-stream probes; contrastive activation addition (CAA)	Sufficiency (improve/correct inferences)	Llama-2, Pythia; 70M–70B; various tuning	Control tasks; PCA-limited probes	Brittleness to prompts; necessity not established
[24]	Spatial/geographic representation	Text	RSA; linear/nonlinear probes; hidden-state interventions	Sufficiency (behavior change); some localization	DeBERTa, GPT-Neo; geospatial task	Layer depth analysis; probe overfitting addressed	Semantic scope is space, not propositional truth/reference
[25]	Propositional logic circuit (A'B)	Text	Activation patching; causal mediation; module-level interventions	Necessity and sufficiency for circuit; modular sub-circuits	Synthetic microtask; Mistral-7B; Gemma-2	QUERY-based patching; sparse decomposable circuit	Task-specific logic; transfer limited
[26]	Othello world model;	Text	Nonlinear probes; activation interventions mapping to	Sufficiency (control)		Latent saliency maps; compare	Linear decodability weak; cir-

	board-state interventions		counterfactual states	next-move probabilities)	Synthetic Othello	linear vs nonlinear	cuit localization limited
[27]	Sylogistic reasoning circuit	Text	Activation patching; targeted ablations; logit/embedding analyses	Necessity and sufficiency; stage-wise circuit	Synthetic syllogisms; GenericsKB variants	Transfer across schemes; contamination by world knowledge	Mechanism susceptible to pretraining biases
[28]	Visual token processing inside LM	Multimodal	Token ablation; logit-lens for visual tokens; attention blocking	Necessity (object info localized); path/time localization	LLaVA (CLIP/Vicuna)	Layerwise refinement toward vocab embeddings	No truth/proposition tests; primarily referential/object info
[29]	Multimodal ToM internal differentiation of perspectives	Multimodal	Attention-head analysis; training-free attention-direction intervention	Sufficiency (enhance exhibited ToM)	GridToM dataset	Confounds in behavioral ToM discussed	Mechanism details limited; truth-conditional mediation not targeted
[30]	Storage/transfer of textual vs visual constraints	Multimodal	Causal information tracing (early MLP/self-attention “causal” for visual constraints)	Localization; some path evidence	LLaVA; Phi-2; VQA-Constraints	Single-/multi-constraint analyses	Intervention details limited; semantics beyond constraint satisfaction not shown
[31]	Circuit probing for intermediate variables (syntax; anaphora)	Text	Trainable binary mask; parameter-level ablation	Necessity for identified circuit components	Synthetic arithmetic; GPT-2 S/M	Conservative sampling; code released	No counterfactual swapping of variables; composition unresolved
[32]	Interventional probing in NLI (monotonicity)	Text	Amnesic/mnesic probing; vector interchange	Mixed: shows pitfalls and limits of interventional probing	NLI natural-logic case study	Strong random baselines; high-dim confound analysis	Demonstrates failure modes; limited positive mediation

Focused comparison: truth-conditional content (truth/belief directions and cones)

Aspect	[2]	[3]	[7,8]	[17]	[18]	[23]	[19]
Representation form	Multidimensional cone subspaces for truth	Approximate linear “truth direction”	Layer-dependent “belief directions”; context-sensitive	Autoencoder-derived “truthful space” + single direction	Multiple per-head “truthfulness” vectors	Rich truthfulness signals in activations; concentrated at specific tokens	Information-state (predicate truth) structured representations
Causal tests	Directional ablation; activation steering flips truth judgments; selectivity	Localized activation shifts near final tokens change model truth behavior	Move premise along direction; measure hypothesis movement; partial mediation	Inference-time edits increase TruthfulQA; opposite edits induce hallucinations	Add steering vectors; adaptive intensity improves truthfulness	Classifier predicts error types; misalignment shows internal truth vs output	Representation edits yield predictable generation changes
Generalization	Across architectures	Probe transfer across datasets; limited model family	Across layers/models/training types; error metrics	Across 13 LLMs; transfers across homologous models	Across many LLaMA-family models	Limited cross-dataset generalization of detectors	Across domains (Alchemy/TextWorld); attributed to pretraining
Selectivity	Preserves unrelated behavior	Not explicitly emphasized	Not assessed	Preserves generation capability	Claimed coverage of diverse hallucinations; selectivity not tightly measured	N/A (detectors)	Edits produce predictable downstream language changes
Limitations	Axes’ semantics partly opaque; scope: simple facts	Truth conflated with belief/commonness; necessity not strong	“One of” causal mediators; deeper pathways not localized	Direction semantics opaque; not circuit-localized	Necessity not shown; probes may capture correlated features	No causal patching; probe-based; selection biases	Primarily probing; limited path-restricted mediation

Key takeaways:

- Multiple lines of evidence support internal variables mediating truth-related behavior, via steering/ablation that flips judgments [2,3,17,18]. However, clear necessity via selective ablations is strongest in [2] (directional ablation), and localization/path specificity is best in [3] (final tokens), with [7,8] showing context-sensitivity and partial mediation.

- Rich but misaligned signals indicate models may internally encode correct answers yet output incorrect ones [23], underscoring the need for mediation tests beyond decodability.
- Earlier dynamic-semantics work shows proposition truth over entities is encoded and editable [19], but without path-restricted causal analysis comparable to [1,4,6,27].

Focused comparison: referential binding, entity tracking, and retrieval

Aspect	[4]	[14,16]	[6]	[1]
Core finding	Binding IDs (BI) mediate entity–attribute binding; subspace structure	Ordering IDs (OI) low-rank subspace acts as BI prototype; causal edits change binding	Three retrieval mechanisms (positional, lexical, reflexive) mixed with context	OIs encode entities/objects/states; visibility look-back encodes observer/observed; multiple look-back mechanisms
Mechanistic methods	Activation patching/interventions; subspace identification	PCA/ICA subspace discovery; activation editing/patching	Interchange interventions; layer-wise patching/ablation; causal abstraction mixture model	Causal mediation tracing; causal abstraction alignment; interchange interventions at variable and feature levels
Causal strength	Necessity (BI affects outputs)	Sufficiency (editing PCs changes binding)	Necessity (specific layer contributions); path specificity; predictive mixture model (~95% agreement)	Necessity and sufficiency; path specificity; selectivity (swap identical surfaces with different OIs)
Localization	Binding subspace; emergent in larger models	Query-side OI subspace; layers not fully mapped	Positional/lexical signals ~layers 12–19; reflexive ~6–12	Residual-stream low-rank subspace co-locates reference; lookback circuits
Robustness	Binding ID distances reflect discriminability	Synthetic tasks; limited to attribute-prediction	Generalizes to long contexts; counterfactual pairs	Generalization checks to larger models/datasets; desiderata-based patching
Limitations	Circuit construction/use left for future work	No circuit identification; synthetic scope; indirect BI–OI interaction	Focused on retrieval; not broader propositional semantics	Synthetic datasets; external validity beyond ToM domain pending

Key takeaways:

- Strong mechanistic evidence that LMs implement referential binding via internal IDs/subspaces [4,14,16], with deeper causal chain and mediation in [1,6].
- Layer-localized retrieval circuits and compositional mixtures suggest distributed causation and context-dependent mechanism blending [6].

Focused comparison: multimodal grounding and cross-modal mediation

Aspect	[10]	[11]	[12]	[13]	[28]	[30]	[29]
Core finding	Binding IDs in VLMs linking image tokens to text referents	Position ID heads (spatial indices) + Feature Retrieval heads	Middle-layer cross-attention heads causally mediate object grounding	“Grounding IDs” induced by cues reduce modality gap and mediate binding	Visual tokens’ object info localized and read out; alignment to text embeddings	Early layers store/transfer visual constraints; causal tracing	Attention heads differentiate perspectives; attention-direction intervention improves ToM
Mechanistic methods	Adapted causal mediation/patching on residual streams	Causal mediation analysis at head outputs; targeted head interventions	Activation patching with SIP/STR corruptions	Attention analyses; layerwise probes; ablations; causal interventions	Token ablation; logit-lens; attention blocking	Causal tracing of information storage/transfer	Attention-head analysis; training-free intervention
Causal strength	Necessity (concept tokens needed; binding vectors mediate)	Necessity (interventions induce binding errors); path specificity	Path specificity; necessity via patching effects concentration	Necessity for binding improvements; causal mediation claims	Necessity (ablation drops >70% accuracy); localization	Localization (early layers “causal”); path evidence	Sufficiency (improve exhibited ToM); causality at attention directions
Grounding scope	Controlled synthetic shapes; standard VLM stack	Spatial binding across image patches	Object detection/suppression roles; cross-attention emphasis	Binding strengthened by external visual/text cues	Object identification; evolution toward text vocabulary	Constraint satisfaction in VQA	Multimodal ToM; internal differentiation of perspectives
Limitations	Focused on binding;	No truth-condition-	Functional roles categorized;	Synthetic; truth-condition-	No proposition-lev-	Intervention details limited; se-	Mechanism details limited; ToM

	truth/proposition not targeted	al analysis; content-independent mechanisms	propositional content not analyzed	al evaluation absent	el tests; referential/object focus	semantic richness unclear	confounds remain
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Key takeaways:

- Multimodal systems show clear causal mediation of referential grounding via spatial indices, cross-attention heads, and binding/grounding IDs [10,11,12,13,28], with localization to specific layers/heads [11,12,28,30].
- Evidence for truth-conditional mediation in multimodal settings remains limited; most work targets referential specificity and constraint satisfaction rather than propositions.

Focused comparison: propositional/world-state representations and reasoning circuits

Aspect	[19]	[20]	[25]	[27]	[26]
Core finding	Dynamic semantics: proposition truth over entities encoded and editable	Propositional probes bind lexical items into predicate-argument propositions via binding subspace	Sparse decomposable circuit for propositional logic microtask	Three-stage syllogistic reasoning circuit; necessary/sufficient via interventions	Internal world model of Othello; board state decodable; interventions alter next moves
Mechanistic methods	Linear probes; representation editing/manipulation	Compositional decoding; binding subspace identification	Activation patching; causal mediation; module interventions	Activation patching; targeted ablations; logit/embedding analyses	Nonlinear probes; activation interventions
Causal strength	Sufficiency (edits cause predictable language changes)	Decoding/monitoring; mediation not detailed	Necessity and sufficiency; modular sub-circuits	Necessity and sufficiency; stage-wise mediation	Sufficiency (control over output probabilities)
Generalization	Across domains; driven by pretraining	From templates to stories; Spanish	Analogous structures in Gemma-2 models	Transfer to common-sense variants; susceptible to knowledge contamination	Synthetic game domain only
Limitations	Limited path-restricted mediation/localization	Interventions not specified; closed-world constraints	Task-specific; broader semantics not shown	Mechanism contaminated by world knowledge/beliefs; not reducible to abstract primitives	Linear decodability weak; circuit-level localization limited

Key takeaways:

- There is converging evidence for internal proposition/world-state representations [19,20,26], with stronger circuit-level causality in synthetic reasoning setups [25,27].
- Bridging these circuit-level methods to open-domain propositional semantics remains a gap.

Methods and validity: frameworks, controls, and pitfalls

- Causal abstraction and interchange interventions:
 - Provide a principled pipeline to align high-level causal models to neural representations and validate via variable-level swaps [5]. Strongly instantiated in belief/binding contexts [1,6].
- Latent causal probing and SCM framing:
 - Formalizes probing with mediation analysis controlling for probe influence, with synthetic demonstrations of necessity/mediation [9].
- Circuit probing and interventional probing:
 - Weight-level circuit discovery with ablations testing causal implication [31], and a case study on NLI that surfaces failure modes and confounds in interventional probing in high dimensions [32].
- Practical pitfalls and controls:
 - Probe overfitting, rank/dimensionality confounds, and correlated features highlighted in [32,24]. Necessity of path-restricted, selective interventions emphasized by multiple works [1,6,27].

Synthesis: where the strongest mechanistic evidence points

- Strongest demonstrations of selective, path-restricted causal mediation for referential binding and belief tracking:

- Binding/Ordering IDs and lookback circuits with interchange/patching and causal abstraction [1,4,6,14].
- Truth-conditional content:
 - Steering and ablation results show sufficiency to flip truth judgments [2,3,17,18], with context-sensitive mediation and partial causal roles [7,8]. Full causal chains and necessity remain less established than in binding.
- Multimodal grounding:
 - Clear causal roles for spatial indices and cross-attention heads mediating referential grounding [10,11,12,13,28,30], but limited evidence on cross-modal propositional truth mediation.
- Reasoning circuits:
 - Synthetic tasks yield necessary-and-sufficient circuit decompositions [25,27]; extending to open-domain semantics is an open challenge.

Overall, the most convincing mechanistic evidence for genuine semantics—operationalized as internal variables that causally mediate reference and, to a lesser extent, truth conditions—comes from studies combining localization, selective interventions, and counterfactual controls in binding and belief-tracking settings [1,4,6]. Truth-related findings increasingly favor multidimensional subspaces (“cones”) and show robust sufficiency via interventions, but require deeper path-level mediation to match the binding evidence [2,3,7,8,17,18].

Timeline

Overview: Phases in the emergence of mechanistic evidence for semantics/intentionality

- 2021–2022: Foundations for causal analysis of meaning in neural networks
 - Formal causal-abstraction pipeline and interchange interventions introduced; first alignment of high-level semantic variables with network internals on quantified NLI, arguing beyond decodability toward causal mediation [5].
 - Early dynamic-semantics probing: internal states decode proposition truth in synthetic grounded transcripts; initial edits/manipulations suggest influence on generation, though primarily via probing with limited causal localization [19].
 - Unsupervised discovery of “truth” directions (CCS) in latent space, framing latent knowledge vs outputs; strong behavioral decoding without circuit localization [22].
 - Synthetic microworlds as testbeds: Othello-GPT establishes non-linear decodability of full world state and activation interventions that control next-move probabilities; a template for closed-world latent state evaluation [26].
- 2023: First explicit mechanisms for reference and truth geometry
 - Binding IDs: large LMs attach explicit binding-ID vectors to entities and attributes; causal interventions show binding-IDs have behavioral consequences; introduces continuous subspace structure for binding and size-dependence of emergence [4].
 - Geometry of truth: emergent linear separability of true/false statements; mass-mean probes transfer across datasets; localized activation interventions flip truth judgments—early causal evidence for a “truth direction” near statement end [3].
 - Methodological deepening around interventions: weight-masked “circuit probing” that localizes variable-specific subnetworks and ablates them to test causal roles on linguistic variables (number, reflexives) [31]; warnings and techniques for interventional probing in high-dimensional NLI (amnesic/mnemonic limits, redundancy) [32].
 - Early spatial/causal representational claims with interventions (geographic space) broaden the semantic scope beyond linguistics [24].
- 2024: From decodability to causal mediation and compositional propositions; early multimodal grounding
 - Propositional probes: compositional decoding of predicate–argument propositions via binding subspaces; supports a latent world model that can diverge from outputs; groundwork for linking binding IDs to propositional content, though limited causal intervention details [20].
 - Logical-reasoning circuits: activation patching and ablations yield necessary-and-sufficient circuits for syllogisms and propositional microtasks in both small trained and pretrained LLMs; reveals modular subcircuits and query-triggered reasoning chains [27,25].
 - Truth/Belief directions at scale: broader studies show context-sensitive, layer-dependent “belief”/truth directions; small causal manipulations move related hypotheses along the same directions, indicating partial mediation in in-context inference [8,7].
 - Truthfulness steering: inference-time editing along learned directions improves TruthfulQA and reduces hallucinations across models (autoencoder split or adaptive steering)—strong behavioral leverage with limited circuit-level mediation [17,18].
 - Internal belief/ToM representations: linear probes and activation edits show model-size and fine-tuning effects, brittleness to prompts, and steerable belief variables in attention-head spaces [21,15].
 - VLM mechanistic pipelines: NOTICE introduces semantically controlled image/text corruptions and cross-modal activation patching to identify causal cross-attention heads; ablations localize object/feature mediation in BLIP/LLaVA [12]. Additional analyses map how CLIP patches flow through LLMs and where object information is read out [28].

- Multimodal causal tracing extends information flow analyses to MLLMs on constraint-specified VQA, identifying early layers as causal for visual constraints, albeit with limited intervention detail [30].
- 2025: Toward richer causal schemas for belief/reference and higher-fidelity truth representations; multimodal binding mechanisms
 - Causal lookbacks and belief tracking: causal mediation plus causal-abstraction alignment reveals low-rank “Ordering IDs” that co-locate reference, and distinct lookback mechanisms (binding, answer, visibility) that mediate who-sees-what beliefs; validated by selective interchange interventions across models/datasets [1]. Follow-up shows competitive synergy between positional, lexical, and reflexive retrieval mechanisms, with layer-localized mediation and a high-fidelity mixture model predictive of logits [6].
 - Beyond directions: cones for propositional truth—truth encoded as multidimensional cones rather than single directions; directional ablations and steering along cone axes flip truth-related behavior while preserving unrelated capabilities across architectures [2].
 - Multimodal binding: Binding-ID style causal analyses adapted to VLMs show patch-level concept carriers and cross-modal Binding IDs linking image tokens to text referents [10]; attention-head-level position IDs and retrieval heads demonstrate content-independent spatial indexing whose targeted disruptions cause binding failures [11]; “Grounding IDs” induced by external cues causally mediate partition-specific binding and reduce hallucinations [13].
 - Multimodal ToM: new GridToM dataset and attention-head–based interventions suggest internal separation of agents’ perspectives in MLLMs and training-free adjustments that enhance exhibited ToM [29].

Trends and methodological shifts

- From probes to causal mediation and path-level alignment
 - Early reliance on linear/nonlinear probes [19,26] shifts toward causal tests: activation patching, targeted ablations, interchange interventions, and causal scrubbing/mediation analyses [5,3,27,25,12,1,6,10,11].
 - Causal abstraction emerges as a unifying framework for aligning high-level semantic variables (logical relations, belief states) to internal subspaces/features, validated via selective interchange [5,1].
- From linear “directions” to low-rank subspaces and cones
 - Initial success with linear truth/belief directions [3,22,8] evolves into low-rank subspaces for binding/ordering IDs [4,14,20] and a cone-based geometry for truth that better captures multifaceted truth signals and improves selectivity/generalization [2].
 - Increasing emphasis on disentanglement/selectivity: necessity/sufficiency shown by directional ablations and minimal-path patching, plus robustness to preserve unrelated behaviors [2,27,25,1,6].
- From entity binding to compositional propositions and belief dynamics
 - Binding IDs and ordering subspaces provide the backbone for reference tracking [4,14,6]; propositional probes move toward predicate–argument truth conditions [20]; belief/ToM variables become steerable and causally implicated in inference [15,21,1].
 - Emerging decomposition: distinct mechanisms for retrieval (positional/lexical/reflexive), visibility/observer encoding, and policy layers vs content layers [6,1].
- From text-only to multimodal grounding with causal evidence
 - VLM analyses progress from localization/ablations [28] to principled causal-mediation pipelines with semantically controlled corruptions [12], and to explicit binding/position-ID circuits whose targeted disruption yields referential failures [10,11,13].
 - Cross-modal alignment is operationalized through IDs that tie spatial indices or partition cues to textual references, moving closer to ground-truth referential mediation claims [10,11,13].
- From closed synthetic worlds to broader settings, with controls
 - Synthetic micro-worlds (Othello, Shapes, syllogisms, grid worlds) remain crucial for ground-truth latent-state tests with clean counterfactuals [26,25,12,10,11,9].
 - Growing attention to robustness: paraphrase/context sensitivity of truth/belief directions [8,7,21], and confounds of instruction tuning or redundancy highlighted in interventional probing critiques [32].

Key contributors and research threads

- Causal abstraction and binding/topology of semantics (Geiger, Steinhardt, colleagues)
 - Causal abstraction methodology and interchange interventions [5] underpin many later works.
 - Binding IDs and ordering subspaces in LMs [4,14,16] culminate in compositional propositional probes [20] and, in 2025, in causal lookback/visibility mechanisms for belief tracking with high-level alignment and interchange validation [1], as well as retrieval-mechanism mixing analyses [6].
 - Influence extends to multimodal adaptations of binding-ID paradigms [10,11,13].
- Truth/Belief geometry and steering (Marks, Tegmark; Burns; follow-ups)
 -

Linear truth directions with causal flips [3] build on unsupervised CCS latent-knowledge identification [22]; later extended to context-sensitive belief/truth directions with causal evidence [8,7].

- Practical steering methods for truthfulness and hallucination mitigation (TruthX, ACT) demonstrate behavioral leverage of truth-related subspaces/directions, albeit with lighter mechanistic localization [17,18].
- 2025 “truth cones” generalize linear direction claims to multidimensional mediation with stronger selectivity and cross-architecture transfer [2].
- Logic/reasoning circuits and compositional mechanisms (Freitas and collaborators; Panigrahy et al.)
 - Syllogistic reasoning circuits with necessary/sufficient components via patching/ablations [27]; propositional-logic micro-tasks yield sparse, modular circuits in both small trained models and pretrained LLMs [25].
 - Methodological cross-pollination with circuit-probing and interventional probing underlines the need for robust counterfactual designs and redundancy-aware controls [31,32].
- Multimodal mechanistic grounding (Eickhoff and collaborators; VLM binding groups)
 - NOTICE pipeline establishes semantically meaningful corruptions and cross-modal causal mediation localization [12]; token-level ablations/logit lens map visual information flow [28]; causal tracing of constraints in MLLMs [30].
 - 2025 VLM works identify binding/position/grounding IDs and head archetypes with causal roles in spatial indexing and cross-modal binding [10,11,13].
- Theory of Mind/belief tracking (Zhu et al.; Bulling et al.; Prakash et al.)
 - Steerable belief directions and generalization across tasks [15]; systematic brittleness/finetuning effects and activation edits to improve ToM [21].
 - Causal lookahead and visibility mechanisms encode who-knows-what with Ordering IDs and validated path-level mediation [1]; multimodal ToM dataset and attention-head adjustments extend to MLLMs [29].

Milestones that advanced the intentionality question

- Establishing causal abstraction and interchange as a standard for “representation vs use,” moving beyond decodability [5].
- Demonstrating explicit binding-ID subspaces and causal mediation for in-context reference [4], later extended and decomposed into multiple retrieval/lookback mechanisms with path-level validation [6,1].
- Showing causal control over truth assessments through targeted latent interventions (first as directions [3], then as cones [2]), and identifying context sensitivity and layer specificity of belief/truth mediation [8,7].
- Progressing from propositional decodability [19] to compositional propositional probes that align binding with predicate–argument structure [20], a step toward truth-conditional internal content.
- Extending mechanistic grounding to multimodal systems with controlled corruptions, head-level causal mediation, and explicit binding/position IDs that mediate cross-modal reference [12,10,11,13].
- Isolating necessary-and-sufficient circuits for logical inference with targeted causal tests, showcasing decomposable computation for content-bearing variables [27,25].

Implications and likely future directions

- Toward operational intentionality tests: Expect broader adoption of causal-abstraction alignment with high-level semantic models (entities, propositions, belief states), verified via interchange/path-restricted patching and sufficiency/necessity tests in both text and multimodal settings [5,1,20,10,11].
- From linearity to structured subspaces: Continued move from single directions to low-rank cones/subspaces that capture polarity, negation, modality, and multi-aspect truth signals with selective interventions and minimal collateral damage [2].
- Closing the “representation–policy” gap: More work to separate world-model content from response policies, explaining internal-external misalignments and enabling precise steering without overfitting instruction-tuning artifacts [23,20,32].
- Richer grounding and interaction: Expansion from passive multimodal alignment to interactive/agentive settings where memory, tools, and environment create persistent state variables (belief, goals) that can be localized and causally manipulated—building on belief/lookback mechanisms and long-context registers [1,6].
- Standardized robustness controls: Synthetic micro-worlds with latent-state ground truth plus paraphrase/adversarial batteries to rule out shortcuts and probe hacking, paired with cross-model and cross-modal generalization tests [26,12,25,32].
- Converging toolkits: Community consolidation around causal mediation, interchange interventions, mixture/path models, and feature dictionary methods to decompose distributed causation while preserving selectivity and counterfactual validity [5,27,1,6,12].

Foundational Work

Which papers form the foundational references on this topic?

The below table shows the resources that are most often cited by the relevant papers on this topic. This is measured by the **reference rate**, which is the fraction of relevant papers that cite a resource. Use this table to determine the most important core papers to be familiar with if you want to deeply understand this topic. Some of these core papers may not be directly relevant to the topic, but provide important context.

Ref.	Reference Rate	Title	Cited By These Relevant Papers
[141]	0.48	How Causal Abstraction Underpins Computational Explanation	[6]
[11]	0.47	Visual symbolic mechanisms: Emergent symbol processing in vision language models	[13]
[20]	0.45	Monitoring Latent World States in Language Models with Propositional Probes	[1, 6, 10, 11, 25]
[4]	0.42	How do Language Models Bind Entities in Context?	[1, 6, 10, 11, 13, 14, 16, 20, 25]
[14]	0.39	Representational Analysis of Binding in Language Models	[1, 6, 13]
[5]	0.33	Causal Abstractions of Neural Networks	[1, 3, 4, 6, 9, 14, 20]
[142]	0.32	Locating and Editing Factual Associations in GPT	[1, 3, 4, 9, 11, 14, 16, 25]
[143]	0.31	What's in the Image? A Deep-Dive into the Vision of Vision Language Models	[10, 11]
[10]	0.30	Investigating Mechanisms for In-Context Vision Language Binding	[13]
[28]	0.29	Towards Interpreting Visual Information Processing in Vision-Language Models	[10, 11]
[144]	0.29	Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking	[1, 4, 6, 10, 14, 16, 20]
[19]	0.28	Implicit Representations of Meaning in Neural Language Models	[1, 3, 4, 9, 20, 21, 24]
[26]	0.25	Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task	[3, 4, 9, 14, 16, 20]
[145]	0.23	Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small	[1, 4, 11, 14, 16, 25]
[3]	0.21	The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets	[2, 7, 8, 14, 16, 20]
[146]	0.20	Entity Tracking in Language Models	[1, 4, 10, 20]
[1]	0.19	Language Models use Lookbacks to Track Beliefs	[6]
[147]	0.17	Designing and Interpreting Probes with Control Tasks	[3, 4, 9, 19]
[148]	0.17	Inference-Time Intervention: Eliciting Truthful Answers from a Language Model	[1, 3, 7, 8]
[48]	0.16	Re-evaluating Theory of Mind evaluation in large language models	[1]

Adjacent Work

Which papers cite the same foundational papers as relevant papers?

Use this table to discover related papers on adjacent topics, to gain a broader understanding of the field and help generate ideas for useful new research directions.

Ref.	Adjacency score	Title	References These Foundational Papers
[141]	2.97	How Causal Abstraction Underpins Computational Explanation	[1, 3, 4, 14, 42]
[159]	1.95	How Do Transformers Learn Variable Binding in Symbolic Programs?	[14, 20, 42]
[160]	1.77	Circuit Distillation	[1, 4, 19]
[161]	1.47	Shape Happens: Automatic Feature Manifold Discovery in LLMs via Supervised Multi-Dimensional Scaling	[4, 14]
[162]	1.39	Behavioural vs. Representational Systematicity in End-to-End Models: An Opinionated Survey	[4, 20]

[163]	1.39	From superposition to sparse codes: interpretable representations in neural networks	[4, 20]
[164]	1.35	Follow the Flow: On Information Flow Across Textual Tokens in Text-to-Image Models	[4, 20]
[165]	1.35	Relational Composition in Neural Networks: A Survey and Call to Action	[4, 20]
[166]	1.35	LINEAR MECHANISMS FOR SPATIOTEMPORAL REASONING IN VISION LANGUAGE MODELS	[4, 20]
[167]	1.30	Mechanistic Interpretability for AI Safety - A Review	[4, 19, 42]
[168]	1.30	RAVEL: Evaluating Interpretability Methods on Disentangling Language Model Representations	[4, 19, 42]
[169]	1.22	MIB: A Mechanistic Interpretability Benchmark	[14, 42]
[170]	1.05	Mechanistic Interpretability of Emotion Inference in Large Language Models	[19, 20]
[171]	0.98	Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability	[4, 42]
[172]	0.92	From Emergence to Control: Probing and Modulating Self-Reflection in Language Models	[20, 22]
[25]	0.87	A Implies B: Circuit Analysis in LLMs for Propositional Logical Reasoning	[4, 20]
[173]	0.83	Mechanistic evaluation of Transformers and state space models	[1]
[174]	0.83	Illusion or Algorithm? Investigating Memorization, Emergence, and Symbolic Processing in In-Context Learning	[1]
[97]	0.79	Relational composition during attribute retrieval in GPT is not purely linear	[4, 157]
[175]	0.73	Dense SAE Latents Are Features, Not Bugs	[20]

References

- [1] [Language Models use Lookbacks to Track Beliefs](#)
Nikhil Prakash, ..., and Atticus Geiger. ArXiv, 2025. 2 citations.
100% Topic Match
Demonstrates a mechanistic account of belief-tracking via lookback mechanisms.
Uses causal mediation, causal-abstraction alignment, and interchange/patch interventions on Llama-3-70B-Instruct trained on synthetic two character CausalToM stories to identify low rank Ordering ID (OI) subspaces and lookback read/write patterns that mediate belief answers.
Provides necessity/sufficiency-style interventions (OI swaps, high level/low level patching), IIA effect quantification, and generalization checks to larger models/datasets—strongly relevant because it shows localized, causally tested internal variables that mediate referential and truth conditional belief content.
- [2] [From Directions to Cones: Exploring Multidimensional Representations of Propositional Facts in LLMs](#)
Stanley Yu, ..., and Vasu Sharma. ArXiv, 2025. 1 citations.
100% Topic Match
Demonstrates a multidimensional "concept cone" representation mediating truthfulness in LLMs.
Identifies cone subspaces and uses causal interventions (activation steering, directional ablation/patching) to flip factual responses and test necessity/sufficiency.
Reports cross model generalization and behavioral selectivity; intervention granularity (cone axes, basis ablations) and semantic axis analyses stated — strong mechanistic/causal evidence relevant to intentionality/truth conditional mediation.
- [3] [The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets](#)
Samuel Marks and Max Tegmark. ArXiv, 2023. 277 citations.
100% Topic Match
Demonstrates a linear "truth direction" in LLM activations.
Visualizes representations, trains linear/mass-mean probes that transfer across datasets, and performs forward-pass activation interventions flipping truth judgments.
Causal interventions (activation/mean-shift) support mediation claims, but scope is simple true/false sentences, limited model scales (mainly LLaMA 13B), and remaining confounds (e.g., "truth" vs correlated signals, probe bias underdetermination).
- [4] [How do Language Models Bind Entities in Context?](#)
Jiahai Feng and Jacob Steinhardt. ArXiv, 2023. 52 citations.
100% Topic Match
Demonstrates a causal binding ID mechanism for entity–attribute binding in large LMs.
Uses representational analyses plus causal interventions (activation manipulations/patching) on Pythia and LLaMA to show internal binding ID vectors that drive outputs.
Relevant: provides necessity/sufficiency-style intervention evidence for interpretable, continuous subspace encodings of referents; focuses on larger models and on representation existence (not full circuit decomposition or multimodal/ truth conditional content).
- [5] [Causal Abstractions of Neural Networks](#)
Atticus Geiger, ..., and Christopher Potts. Unknown journal, 2021. 285 citations.
100% Topic Match
Demonstrates a formal causal-abstraction method aligning neural states to interpretable causal variables.
Aligns neural representations to a tree-structured natural-logic model and uses interchange interventions (representation swapping) to test causal equivalence.
Case study: BERT partially realizes MQNLI's compositional causal structure (interchange shows necessity/sufficiency for some variables); contrasts with LSTM; argues this goes beyond probes but requires hypothesized high level model.
- [6] [Mixing Mechanisms: How Language Models Retrieve Bound Entities In-Context](#)
Yoav Gur-Arieh, ..., and Atticus Geiger. ArXiv, 2025. 0 citations.
100% Topic Match
Demonstrates a mechanistic account of how LMs bind and retrieve entities in-context.
Uses counterfactual input pairs and extensive interchange interventions/patching on residual-stream vectors to identify positional, lexical, and reflexive retrieval mechanisms and fit a P/L/R mixture causal model predicting next-token logits (~95% agreement).
Relevant because it performs localized causal interventions across layers and models to show necessity/sufficiency of internal signals for referent retrieval, but scope is largely entity-binding (who/what retrieval) rather than broader truth conditional propositional content or multimodal/agentive grounding.
- [7] [Truth-value judgment in language models: 'truth directions' are context sensitive](#)
Stefan F. Schouten, ..., and Piek Vossen. ArXiv, 2024. 0 citations.
100% Topic Match
Demonstrates that LLM latent-space "truth/belief directions" are context sensitive.
Probes linear truth-value directions across layers, measures consistency/consistency to supporting/negating contexts, and performs representation shifts to test causal mediation.
Finds probes often change with irrelevant context, errors vary by layer/model/training, and intervention results suggest truth directions can be causal mediators—but not fully stable or context-invariant.
- [8] [Truth-value judgment in language models: belief directions are context sensitive](#)
Stefan F. Schouten, ..., and Piek Vossen. ArXiv, 2024. 2 citations.
100% Topic Match
Demonstrates that LLM latent "belief directions" encode context sensitive truth signals.
Tests linear probes (logistic, CCS/variant, MMP) across layers/models and shows probe outputs shift with supporting/contradicting/negated context; performs an intervention that moves a premise along the belief direction and measures corresponding shifts in the hypothesis.
Relevant: provides causal mediation evidence (single direction activation steering) but stops short of full mechanistic circuit/path analyses, has layer/model/data dependencies, and notes limitations—so it's somewhat relevant (causal but limited in scope and depth).
- [9] [Latent Causal Probing: A Formal Perspective on Probing with Causal Models of Data](#)
Charles Jin and Martin Rinard. ArXiv, 2024. 3 citations.
100% Topic Match
Proposes a formal latent-causal probing framework using structural causal models.
Frames probing as testing whether pretrained LMs represent SCM latent variables, adding causal mediation analyses to isolate LM-internal mediation from probe effects.
Evaluated on a synthetic grid-world with known latent states; provides necessity/mediation tests (not just decodability) and probe-confound controls, but uses synthetic tasks rather than large multimodal/agentive LLMs.
- [10] [Investigating Mechanisms for In-Context Vision Language Binding](#)
Darshana Saravanan, ..., and Vineet Gandhi. 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2025. 0 citations.
100% Topic Match
Demonstrates a Binding ID style mechanism for image–text association in VLMs.
Adapts Feng & Steinhardt's causal mediation Binding ID protocol on a synthetic 3D "Shapes" QA task, inspecting residual stream token activations across visual encoder ' projector ' LLM to find linking subspaces/vectors.

Uses controlled synthetic data and token level interventions/analyses showing image patches and textual mentions share distinct binding vectors/subspaces; check whether interventions test necessity/sufficiency and path restricted patching—paper claims causal mediation but may not report exhaustive sufficiency/robustness controls across paraphrases, modalities, or counterfactuals.

[11] [Visual symbolic mechanisms: Emergent symbol processing in vision language models](#)

Rim Assouel, ..., and Taylor Webb. ArXiv, 2025. 0 citations.

100% Topic Match

Demonstrates emergent symbol like spatial indexing mechanisms that mediate visual feature binding in VLMs.

Identifies Position ID and Feature Retrieval attention heads, maps representational progression, and uses representational analyses plus causal mediation and targeted interventions on head outputs.

Relevant for referential/internal mediation of reference (spatial/object identity) with necessity/sufficiency interventions, but does not test truth conditional propositional content, cross modal grounding beyond spatial indices, or agentic/interactive grounding.

[12] [What Do VLMs NOTICE? A Mechanistic Interpretability Pipeline for Noise-free Text-Image Corruption and Evaluation](#)

Michal Golovanevsky, ..., and Carsten Eickhoff. ArXiv, 2024. 10 citations.

100% Topic Match

Introduces NOTICE: a mechanistic interpretability pipeline for causally testing multimodal grounding in VLMs.

Constructs semantic minimal-pair corruptions (SIP/STR) and applies activation patching/causal mediation to BLIP/LLaVA, finding large middle-layer cross attention patching effects.

Relevant because it uses causal interventions (activation patching) on identified attention heads, reports necessity/sufficiency-like effects for image grounding, and warns against noisy corruptions—good match if you prioritize mechanistic, cross modal mediation tests.

[13] [Uncovering Grounding IDs: How External Cues Shape Multi-Modal Binding](#)

Hosein Hasani, ..., and M. Baghshah. ArXiv, 2025. 0 citations.

99% Topic Match

Demonstrates a causal role for latent "Grounding ID" vectors in multimodal referent binding.

Uses synthetic LVM experiments (Qwen2.5-VL + controlled shapes/colors), representation analyses, attention/head tracing, layerwise probes, ablations, and causal interventions to show partition-induced ID vectors align image/text embeddings and mediate binding.

Relevant because it presents mechanistic internal evidence (identifiable subspace/vectors, attention propagation) plus causal mediation tests for referential grounding and hallucination reduction—but check intervention rigor (specifics of path-restricted patching, sufficiency tests, OOD/paraphrase robustness, and whether interventions selectively flip truth-conditional content rather than global performance).

[14] [Representational Analysis of Binding in Language Models](#)

Qin Dai, ..., and Kentaro Inui. Unknown journal, 2024. 4 citations.

99% Topic Match

Demonstrates a localized, low-rank subspace encoding Binding/Ordering IDs in LM hidden states.

Identifies the subspace via dimensionality reduction (PCA/ICA) and causally alters binding behavior by editing/patching principal-component directions in query activations, producing predictable retrieval swaps on synthetic entity tracking tasks.

Relevant limits: uses synthetic datasets, inspects only query activations (not context), infers BI changes from outputs (no direct circuit identification or path-restricted mediation), and tests several ~7–8B pretrained models.

[15] [Language Models Represent Beliefs of Self and Others](#)

Wentao Zhu, ..., and Yizhou Wang. ArXiv, 2024. 13 citations.

96% Topic Match

Demonstrates a linear, causally effective internal belief representation in an LLM.

Trains linear probes on attention head activations in Mistral 7B Instruct, identifies consistent belief directions, and manipulates those vectors to strongly change ToM task outputs.

Evidence includes gradient linking to input tokens and cross task generalization, but limited to one model family, attention space probes, and social reasoning tasks (necessity/sufficiency shown via vector steering, not full path restricted circuit decomposition).

[16] [Representational Analysis of Binding in Large Language Models](#)

Qin Dai, ..., and Kentaro Inui. ArXiv, 2024. 0 citations.

96% Topic Match

Demonstrates a low-rank hidden state subspace (Binding ID prototype) that encodes entity–attribute binding.

Identifies this subspace via PCA on activations and links it causally to binding behavior as the BI prototype.

Focuses on representational localization (low rank subspace) for entity tracking; check whether they run causal interventions (patching/ablation/interchange) and robustness/paraphrase controls—if absent, relevance is limited to correlational localization rather than full necessity/sufficiency evidence.

[17] [TruthX: Alleviating Hallucinations by Editing Large Language Models in Truthful Space](#)

Shaolei Zhang, ..., and Yang Feng. Unknown journal, 2024. 72 citations.

94% Topic Match

Demonstrates an inference-time intervention that edits internal activations to increase LLM truthfulness.

Trains an autoencoder to split model representations into semantic vs "truthful" latent spaces, finds a single truthful editing direction via contrastive learning, and shifts activations at inference to boost TruthfulQA performance (~+20% across 13 LLMs).

Relevant caveats: evidence is interventional (activation editing) showing necessity/sufficiency of one direction for output truthfulness, but paper may not provide full causal-path decomposition, rigorous path-restricted patching, or grounding tests across modalities/agentic settings — check whether controls for probe-hacking, counterfactual minimal pairs, and cross-context robustness are reported.

[18] [Adaptive Activation Steering: A Tuning-Free LLM Truthfulness Improvement Method for Diverse Hallucinations Categories](#)

Tianlong Wang, ..., and Liantao Ma. Proceedings of the ACM on Web Conference 2025, 2024. 43 citations.

91% Topic Match

Demonstrates a tuning-free method (Adaptive Activation Steering, ACT) to steer LLM activations toward "truthful" outputs.

Trains linear probes on last-token head activations to locate truthfulness directions, constructs diverse steering vectors, and adds them adaptively at inference to improve truthfulness across LLaMA-family models and sizes.

Relevant caveats: evidence is decodability+steering at head activation level (inference-time add-on) showing behavioral gains; but paper tests linear directional interventions (sufficiency) without detailed path restricted mediation, minimal path/ablation analyses, or thorough counterfactual grounding across modalities/agentic settings.

[19] [Implicit Representations of Meaning in Neural Language Models](#)

Belinda Z. Li, ..., and Jacob Andreas. Unknown journal, 2021. 190 citations.

87% Topic Match

Demonstrates that pretrained transformers encode dynamic, truth conditional entity/state representations.

Probes BART and T5 with linear decoders on synthetic/text game domains, edits internal representations, and shows predictable downstream generation changes.

Uses probing plus representation editing (no extensive causal path patching); evidence for localized, manipulable proposition encodings but limited causal mediation tests and mostly text only grounding.

[20] [Monitoring Latent World States in Language Models with Propositional Probes](#)

Jiahai Feng, ..., and Jacob Steinhardt. ArXiv, 2024. 16 citations.

81% Topic Match

Demonstrates a compositional method to decode propositional/world-state content from LM activations.

Uses "propositional probes": extract lexical token vectors, identify a binding subspace, and bind tokens into predicate–argument propositions; validated in closed-world templates, stories, and translations.

Relevant: provides internal representational evidence for latent propositional content and monitoring, but lacks reported causal-intervention tests (necessity/sufficiency, patching, ablations) tying decoded probes to model behavior.

[21] [Brittle Minds, Fixable Activations: Understanding Belief Representations in Language Models](#)

Matteo Bortoletto, ..., and Andreas Bulling. ArXiv, 2024. 3 citations.

75% Topic Match

Analyzes internal belief (Theory of Mind) representations in LMs.

Trains layerwise linear probes on residual-stream activations across models/scales, uses control tasks/PC tests, and applies targeted activation edits (CAA) to steer ToM outputs.

Relevant: provides evidence of structured, low dimensional, causally manipulable belief features, but relies mostly on linear probing + activation editing (not path restricted patching or full mediation/path analyses); brittleness to prompts and focus on residual stream implies partial mechanistic evidence rather than complete necessity/sufficiency proofs for truth conditional intentionality.

[22] [Discovering Latent Knowledge in Language Models Without Supervision](#)

Collin Burns, ..., and J. Steinhardt. ArXiv, 2022. 453 citations.

64% Topic Match

Demonstrates an unsupervised method to extract yes/no "latent knowledge" from LLM activations.

Uses Contrast Consistent Search (CCS) to find an activation-space direction satisfying logical consistency (statement vs negation) and map activations to truth labels, improving zero shot accuracy and reducing prompt sensitivity.

Relevant caveats: shows decodability of truth-related content but lacks mechanistic localization or causal mediation tests (no patching/ablations/interchange); depends on existence of a separable direction and does not probe deception/agentive grounding.

[23] [LLMs Know More Than They Show: On the Intrinsic Representation of LLM Hallucinations](#)

Hadas Orgad, ..., and Yonatan Belinkov. ArXiv, 2024. 98 citations.

58% Topic Match

Shows that LLM internal states encode rich, concentrated truthfulness signals.

Uses supervised probes/classifiers on token-level activations from greedy-decoded generations to detect truthfulness and error types, improving detection by exploiting concentrated tokens.

No reported causal interventions or circuit level localization; detectors fail to generalize across datasets and models can encode correct answers without producing them, so mechanistic necessity/sufficiency for semantics remains unestablished.

[24] [More than Correlation: Do Large Language Models Learn Causal Representations of Space?](#)

Yida Chen, ..., and Xiaohan Zhao. ArXiv, 2023. 7 citations.

49% Topic Match

Demonstrates a causal role for putative spatial representations in LLM behavior.

Uses RSA plus linear/nonlinear probes to identify spatial subspaces in DeBERTa and GPT Neo, then intervenes on hidden states to show changes in next token prediction and a geospatial downstream task.

Intervention is at intermediate hidden states (citing prior intervention methods); provides probe robustness discussion (probe saturation across layers, nonlinear probe gains), but is limited to space (geospatial) and models tested—check whether interventions are path-restricted/sufficiency tests and whether confounds were fully ruled out.

[25] [A Implies B: Circuit Analysis in LLMs for Propositional Logical Reasoning](#)

Guan Zhe Hong, ..., and Rina Panigrahy. ArXiv, 2024. 0 citations.

46% Topic Match

Demonstrates a sparse, decomposable reasoning circuit for propositional logic in LLMs.

Uses a synthetic propositional-logic microtask plus activation patching/causal mediation and multi-level interventions to locate residual-stream, attention-head, and MLP components that change logits on counterfactual prompts.

Relevant because it applies causal interventions (activation/patching, QUERY-based patching) and circuit decomposition across small transformers and larger pretrained models (Mistral-7B, Gemma-2-9B/27B), but focuses on synthetic logical microtasks rather than naturalistic referential/truth-conditional language grounding.

[26] [Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task](#)

Kenneth Li, ..., and M. Wattenberg. ArXiv, 2022. 324 citations.

37% Topic Match

Demonstrates an emergent internal world-model for a synthetic board-game task.

Trains a GPT-style model on Othello transcripts, decodes full board state with a nonlinear probe, and uses activation interventions to shift next-move probabilities.

Relevant: provides mechanistic-ish causal tests (activation edits) showing some necessity/sufficiency, but relies on nonlinear probes, a toy synthetic domain, and interventions not proven path-restricted—limits generalizability to large LLMs, multimodal grounding, and strict truth-conditional/reference mediation.

[27] [Reasoning Circuits in Language Models: A Mechanistic Interpretation of Syllogistic Inference](#)

Geonhee Kim, ..., and André Freitas. Unknown journal, 2024. 11 citations.

29% Topic Match

Demonstrates a mechanistic circuit for syllogistic inference in autoregressive LMs.

Uses synthetic syllogistic tasks plus naturalistic instantiations, activation patching, targeted ablations, and logit lens analyses to show a three stage attention/residual circuit is necessary and sufficient for valid conclusions.

Relevant because it provides causal interventions, localization to attention/residual pathways, and generalization tests across schemes/models — but notes limitations (e60% accuracy regimes, contamination by world knowledge heads, not reducible to abstract logical primitives).

[28] [Towards Interpreting Visual Information Processing in Vision-Language Models](#)

Clement Neo, ..., and Fazl Barez. ArXiv, 2024. 39 citations.

24% Topic Match

Analyzes how visual tokens are processed and localized inside a VLM's language-model component.

Uses token-level ablations, attention-blocking, and logit lens on LLaVA (CLIP ViT ' adapter ' Vicuna-13B) to show token localization, layerwise refinement toward vocabulary-aligned representations, and readout at final token positions.

Relevant: provides mechanistic internal analyses and causal-style ablations for referent localization in multimodal context, but lacks targeted interchange/patching, path-restricted mediation, or truth conditional/propositional intervention tests—so it partly meets but does not fully satisfy the paper's strict criteria for demonstrating genuine semantics/intentionality.

[29] [From Black Boxes to Transparent Minds: Evaluating and Enhancing the Theory of Mind in Multimodal Large Language Models](#)

Xinyang Li, ..., and Huimin Ma. ArXiv, 2025. 0 citations.

18% Topic Match

Demonstrates an interpretability-driven evaluation of Theory of Mind in multimodal LLMs.

Builds GridToM (multimodal belief tasks), analyzes attention heads that differentiate agent perspectives, and applies a lightweight, training free attention-direction intervention to boost ToM behavior.

Mechanistic evidence is limited to attention-head readouts and a directional adjustment intervention—look for whether they run necessity/sufficiency causal tests (patching/ablation/interchange) and robustness controls; behavioral confounds (hallucination, scenario understanding) are discussed but full causal mediation of truth-conditional content or cross-modal interchange interventions are not clearly reported.

[30] [Understanding Information Storage and Transfer in Multi-modal Large Language Models](#)

Samyadeep Basu, ..., and Daniela Massiceti. ArXiv, 2024. 28 citations.

16% Topic Match

Extends causal information tracing analyses to multimodal LLMs (MLLMs).

Introduces VQA Constraints dataset and a MULTIMODALCAUSALTRACE method to trace where visual/textual constraint information is stored and transferred in LLaVa and Phi 2.

Relevant if you want layerwise/mechanistic localization of referential constraint signals in MLLMs, but paper (as described) lacks detailed causal interventions (e.g., weight/activation patching, interchange, path restricted mediation) and so may not fully test necessity/sufficiency for intentionality.

[31] [Uncovering Intermediate Variables in Transformers using Circuit Probing](#)

Michael A. Lepori, ..., and Ellie Pavlick. ArXiv, 2023. 9 citations.

14% Topic Match

Demonstrates a method (circuit probing) to localize weight level circuits tied to hypothesized intermediate variables.

Learns binary masks over Transformer weights, isolates parameter subsets, then performs targeted parameter ablations and output level classification to test causal involvement.

Relevant: provides necessity-style ablations (better than decoding alone) and tracks circuit development, but does not perform counterfactual/state swap interventions or show sufficiency via forced internal variable insertion.

[32] [Interventional Probing in High Dimensions: An NLI Case Study](#)

Julia Rozanova, ..., and André Freitas. Unknown journal, 2023. 8 citations.

11% Topic Match

Analyzes the causal role of probe-extracted semantic features in NLI model representations.

Applies amnesic probing (remove probe directions), introduces mnestic probing (keep only probe directions), and vector swap interventions, measuring effects on NLI (natural-logic/monotonicity) labels.

Finds high-dimensional redundancy, nonlinear encoding, and probe confounds limit causal conclusions; recommends stronger baselines, controls, and notes interventions may not establish necessity/sufficiency for truth-conditional semantics.

[33] [Causal Interventions on Causal Paths: Mapping GPT-2's Reasoning From Syntax to Semantics](#)

Isabelle Lee, ..., and Dani Yogatama. ArXiv, 2024. 0 citations.

9% Topic Match

Abstract: While interpretability research has shed light on some internal algorithms utilized by transformer-based LLMs, reasoning in natural language, with its deep contextuality and ambiguity, defies easy categorization. As a result, formulating clear and motivating questions for circuit analysis that rely on well-defined in-domain and out-of-domain examples required for causal interventions is challenging. Although significant work has investigated circuits for specific tasks, such as indirect object identification (IOI), deciphering natural language reasoning through circuits remains difficult due to its inherent complexity. In this work, we take initial steps to characterize causal reasoning in LLMs by analyzing clear-cut cause-and-effect sentences like "I opened..."

[34] [Does ChatGPT Have a Mind?](#)

Simon Goldstein and B. Levinstein. ArXiv, 2024. 10 citations.

7% Topic Match

Abstract: This paper examines the question of whether Large Language Models (LLMs) like ChatGPT possess minds, focusing specifically on whether they have a genuine folk psychology encompassing beliefs, desires, and intentions. We approach this question by investigating two key aspects: internal representations and dispositions to act. First, we survey various philosophical theories of representation, including informational, causal, structural, and teleosemantic accounts, arguing that LLMs satisfy key conditions proposed by each. We draw on recent interpretability research in machine learning to support these claims. Second, we explore whether LLMs exhibit robust dispositions to perform actions, a necessary component of folk psychology. We...

[35] [Can structural correspondences ground real world representational content in Large Language Models?](#)

Iwan Williams. ArXiv, 2025. 0 citations.

5% Topic Match

Abstract: Large Language Models (LLMs) such as GPT-4 produce compelling responses to a wide range of prompts. But their representational capacities are uncertain. Many LLMs have no direct contact with extra-linguistic reality: their inputs, outputs and training data consist solely of text, raising the questions (1) can LLMs represent anything and (2) if so, what? In this paper, I explore what it would take to answer these questions according to a structural-correspondence based account of representation, and make an initial survey of this evidence. I argue that the mere existence of structural correspondences between LLMs and worldly entities is insufficient to...

[36] [The Vector Grounding Problem](#)

Dimitri Coelho Mollo and Raphael Milliere. ArXiv, 2023. 31 citations.

4% Topic Match

Abstract: The remarkable performance of large language models (LLMs) on complex linguistic tasks has sparked debate about their capabilities. Unlike humans, these models learn language solely from textual data without directly interacting with the world. Yet they generate seemingly meaningful text on diverse topics. This achievement has renewed interest in the classical 'Symbol Grounding Problem' -- the question of whether the internal representations and outputs of symbolic AI systems can possess intrinsic meaning that is not parasitic on external interpretation. Although modern LLMs compute over vectors rather than symbols, an analogous problem arises for these systems, which we call the Vector...

[37] [Detecting \(Un\)answerability in Large Language Models with Linear Directions](#)

Maor Juliet Lavi, ..., and Mor Geva. ArXiv, 2025. 0 citations.

4% Topic Match

Abstract: Large language models (LLMs) often respond confidently to questions even when they lack the necessary information, leading to hallucinated answers. In this work, we study the problem of (un)answerability detection, focusing on extractive question answering (QA) where the model should determine if a passage contains sufficient information to answer a given question. We propose a simple approach for identifying a direction in the model's activation space that captures unanswerability and uses it for classification. This direction is selected by applying activation additions during inference and measuring their impact on the model's abstention behavior. We show that projecting hidden activations onto...

[38] [On Mechanistic Circuits for Extractive Question-Answering](#)

Samyadeep Basu, ..., and Varun Manjunatha. ArXiv, 2025. 0 citations.

3% Topic Match

Abstract: Large language models are increasingly used to process documents and facilitate question-answering on them. In our paper, we extract mechanistic circuits for this real-world language modeling task: context-augmented language modeling for extractive question-answering (QA) tasks and understand the potential benefits of circuits towards downstream applications such as data attribution to context information. We extract circuits as a function of internal model components (e.g., attention heads, MLPs) using causal mediation analysis techniques. Leveraging the extracted circuits, we first understand the interplay between the model's usage of parametric memory and retrieved context towards a better mechanistic understanding of context-augmented language models. We...

[39] [Caught in the Act: a mechanistic approach to detecting deception](#)

Gerard Boxo, ..., and Shivam Raval. ArXiv, 2025. 0 citations.

2% Topic Match

Abstract: Sophisticated instrumentation for AI systems might have indicators that signal misalignment from human values, not unlike a "check engine" light in cars. One such indicator of misalignment is deceptiveness in generated responses. Future AI instrumentation may have the ability to detect when an LLM generates deceptive responses while reasoning about seemingly plausible but incorrect answers to factual questions. In this work, we demonstrate that linear probes on LLMs internal activations can detect deception in their responses with extremely high accuracy. Our probes reach a maximum of greater than 90% accuracy in distinguishing between deceptive and non-deceptive arguments generated by llama and qwen...

[40] [Position-aware Automatic Circuit Discovery](#)

Tal Haklay, ..., and Yonatan Belinkov. Unknown journal, 2025. 3 citations.

2% Topic Match

Abstract: A widely used strategy to discover and understand language model mechanisms is circuit analysis. A circuit is a minimal subgraph of a model's computation graph that executes a specific task. We identify a gap in existing circuit discovery methods: they assume circuits are position-invariant, treating model components

as equally relevant across input positions. This limits their ability to capture cross-positional interactions or mechanisms that vary across positions. To address this gap, we propose two improvements to incorporate positionality into circuits, even on tasks containing variable-length examples. First, we extend edge attribution patching, a gradient-based method for circuit discovery, to differentiate...

[41] [LLM Knowledge is Brittle: Truthfulness Representations Rely on Superficial Resemblance](#)

Patrick Haller, ..., and Samuel J. Bell. ArXiv, 2025. 0 citations.

2% Topic Match

Abstract: For Large Language Models (LLMs) to be reliable, they must learn robust knowledge that can be generally applied in diverse settings -- often unlike those seen during training. Yet, extensive research has shown that LLM performance can be brittle, with models exhibiting excessive sensitivity to trivial input variations. In this work, we explore whether this brittleness is a direct result of unstable internal knowledge representations. To explore this question, we build on previous work showing that LLM representations encode statement truthfulness -- i.e., true, factual statements can be easily separated from false, inaccurate ones. Specifically, we test the robustness of...

[42] [Discovering Variable Binding Circuitry with Desiderata](#)

Xander Davies, ..., and David Bau. ArXiv, 2023. 18 citations.

2% Topic Match

Abstract: Recent work has shown that computation in language models may be human-understandable, with successful efforts to localize and intervene on both single-unit features and input-output circuits. Here, we introduce an approach which extends causal mediation experiments to automatically identify model components responsible for performing a specific subtask by solely specifying a set of `{desiderata}`, or causal attributes of the model components executing that subtask. As a proof of concept, we apply our method to automatically discover shared `{variable binding circuitry}` in LLaMA-13B, which retrieves variable values for multiple arithmetic tasks. Our method successfully localizes variable binding to only 9 attention...

[43] [An explainable transformer circuit for compositional generalization](#)

Cheng Tang, ..., and Mehrdad Jazayeri. ArXiv, 2025. 7 citations.

1% Topic Match

Abstract: Compositional generalization-the systematic combination of known components into novel structures-remains a core challenge in cognitive science and machine learning. Although transformer-based large language models can exhibit strong performance on certain compositional tasks, the underlying mechanisms driving these abilities remain opaque, calling into question their interpretability. In this work, we identify and mechanistically interpret the circuit responsible for compositional induction in a compact transformer. Using causal ablations, we validate the circuit and formalize its operation using a program-like description. We further demonstrate that this mechanistic understanding enables precise activation edits to steer the model's behavior predictably. Our findings advance the understanding...

[44] [Personas as a Way to Model Truthfulness in Language Models](#)

Nitish Joshi, ..., and He He. ArXiv, 2023. 39 citations.

1% Topic Match

Abstract: Large language models (LLMs) are trained on vast amounts of text from the internet, which contains both factual and misleading information about the world. While unintuitive from a classic view of LMs, recent work has shown that the truth value of a statement can be elicited from the model's representations. This paper presents an explanation for why LMs appear to know the truth despite not being trained with truth labels. We hypothesize that the pretraining data is generated by groups of (un)truthful agents whose outputs share common features, and they form a (un)truthful persona. By training on this data, LMs...

[45] [No Answer Needed: Predicting LLM Answer Accuracy from Question-Only Linear Probes](#)

Iv'an Vicente Moreno Cencerrado, ..., and Lorenzo Pacchiardi. ArXiv, 2025. 1 citations.

1% Topic Match

Abstract: Do large language models (LLMs) anticipate when they will answer correctly? To study this, we extract activations after a question is read but before any tokens are generated, and train linear probes to predict whether the model's forthcoming answer will be correct. Across three open-source model families ranging from 7 to 70 billion parameters, projections on this "in-advance correctness direction" trained on generic trivia questions predict success in distribution and on diverse out-of-distribution knowledge datasets, outperforming black-box baselines and verbalised predicted confidence. Predictive power saturates in intermediate layers, suggesting that self-assessment emerges mid-computation. Notably, generalisation falters on questions requiring mathematical reasoning. Moreover,...

[46] [Emergent Symbolic Mechanisms Support Abstract Reasoning in Large Language Models](#)

Yukang Yang, ..., and Taylor Webb. ArXiv, 2025. 10 citations.

1% Topic Match

Abstract: Many recent studies have found evidence for emergent reasoning capabilities in large language models (LLMs), but debate persists concerning the robustness of these capabilities, and the extent to which they depend on structured reasoning mechanisms. To shed light on these issues, we study the internal mechanisms that support abstract reasoning in LLMs. We identify an emergent symbolic architecture that implements abstract reasoning via a series of three computations. In early layers, symbol abstraction heads convert input tokens to abstract variables based on the relations between those tokens. In intermediate layers, symbolic induction heads perform sequence induction over these abstract variables...

[47] [Operationalising Representation in Natural Language Processing](#)

J. Harding. ArXiv, 2023. 15 citations.

1% Topic Match

Abstract: Despite its centrality in the philosophy of cognitive science, there has been little prior philosophical work engaging with the notion of representation in contemporary NLP practice. This paper attempts to fill that lacuna: drawing on ideas from cognitive science, I introduce a framework for evaluating the representational claims made about components of neural NLP models, proposing three criteria with which to evaluate whether a component of a model represents a property and operationalising these criteria using probing classifiers, a popular analysis technique in NLP (and deep learning more broadly). The project of operationalising a philosophically-informed notion of representation should be...

[48] [Re-evaluating Theory of Mind evaluation in large language models](#)

Jennifer Hu, ..., and Tomer D. Ullman. Philosophical transactions of the Royal Society of London. Series B, Biological sciences, 2025. 6 citations.

1% Topic Match

Abstract: The question of whether large language models (LLMs) possess Theory of Mind (ToM)-often defined as the ability to reason about others' mental states-has sparked significant scientific and public interest. However, the evidence as to whether LLMs possess ToM is mixed, and the recent growth in evaluations has not resulted in a convergence. Here, we take inspiration from cognitive science to re-evaluate the state of ToM evaluation in LLMs. We argue that a major reason for the disagreement on whether LLMs have ToM is a lack of clarity on whether models should be expected to match human behaviours, or the computations...

[49] [Towards Uncovering How Large Language Model Works: An Explainability Perspective](#)

Haiyan Zhao, ..., and Mengnan Du. ArXiv, 2024. 21 citations.

1% Topic Match

Abstract: Large language models (LLMs) have led to breakthroughs in language tasks, yet the internal mechanisms that enable their remarkable generalization and reasoning abilities remain opaque. This lack of transparency presents challenges such as hallucinations, toxicity, and misalignment with human values, hindering the safe and beneficial deployment of LLMs. This paper aims to uncover the mechanisms underlying LLM functionality through the lens of explainability. First, we review how knowledge is architecturally composed within LLMs and encoded in their internal parameters via mechanistic interpretability techniques. Then, we summarize how knowledge is embedded in LLM representations by leveraging probing techniques and representation engineering...

[50] [Towards Controllable and Explainable Text Generation via Causal Intervention in LLMs](#)

Jie Qiu, ..., and Wenhao Kang. Electronics, 2025. 1 citations.

1% Topic Match

Abstract: Large Language Models (LLMs) excel in diverse text generation tasks but still face limited controllability, opaque decision processes, and frequent hallucinations. This paper presents a structural causal intervention framework that models input-hidden-output dependencies through a structural causal model and performs targeted interventions on hidden representations. By combining counterfactual sample construction with contrastive training, our method enables precise control of style, sentiment, and factual consistency while providing explicit causal explanations for output changes. Experiments on three representative tasks

demonstrate consistent and substantial improvements: style transfer accuracy reaches 92.3% (+7–14 percentage points over strong baselines), sentiment-controlled generation achieves 90.1% accuracy (+1.3–10.9 points),...

[51] [Probing the Geometry of Truth: Consistency and Generalization of Truth Directions in LLMs Across Logical Transformations and Question Answering Tasks](#)

Yuntai Bao, ..., and Jianwei Yin. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: Large language models (LLMs) are trained on extensive datasets that encapsulate substantial world knowledge. However, their outputs often include confidently stated inaccuracies. Earlier works suggest that LLMs encode truthfulness as a distinct linear feature, termed the "truth direction", which can classify truthfulness reliably. We address several open questions about the truth direction: (i) whether LLMs universally exhibit consistent truth directions; (ii) whether sophisticated probing techniques are necessary to identify truth directions; and (iii) how the truth direction generalizes across diverse contexts. Our findings reveal that not all LLMs exhibit consistent truth directions, with stronger representations observed in more capable models, particularly...

[52] [Have Faith in Faithfulness: Going Beyond Circuit Overlap When Finding Model Mechanisms](#)

Michael Hanna, ..., and Yonatan Belinkov. ArXiv, 2024. 52 citations.

0% Topic Match

Abstract: Many recent language model (LM) interpretability studies have adopted the circuits framework, which aims to find the minimal computational subgraph, or circuit, that explains LM behavior on a given task. Most studies determine which edges belong in a LM's circuit by performing causal interventions on each edge independently, but this scales poorly with model size. Edge attribution patching (EAP), gradient-based approximation to interventions, has emerged as a scalable but imperfect solution to this problem. In this paper, we introduce a new method - EAP with integrated gradients (EAP-IG) - that aims to better maintain a core property of circuits: faithfulness...

[53] [Unveiling Theory of Mind in Large Language Models: A Parallel to Single Neurons in the Human Brain](#)

Mohsen Jamali, ..., and Jing Cai. ArXiv, 2023. 20 citations.

0% Topic Match

Abstract: With their recent development, large language models (LLMs) have been found to exhibit a certain level of Theory of Mind (ToM), a complex cognitive capacity that is related to our conscious mind and that allows us to infer another's beliefs and perspective. While human ToM capabilities are believed to derive from the neural activity of a broadly interconnected brain network, including that of dorsal medial prefrontal cortex (dmPFC) neurons, the precise processes underlying LLM's capacity for ToM or their similarities with that of humans remains largely unknown. In this study, we drew inspiration from the dmPFC neurons subserving human ToM...

[54] [Testing theory of mind in large language models and humans](#)

James W. A. Strachan, ..., and Cristina Becchio. Nature Human Behaviour, 2024. 216 citations.

0% Topic Match

Abstract: At the core of what defines us as humans is the concept of theory of mind: the ability to track other people's mental states. The recent development of large language models (LLMs) such as ChatGPT has led to intense debate about the possibility that these models exhibit behaviour that is indistinguishable from human behaviour in theory of mind tasks. Here we compare human and LLM performance on a comprehensive battery of measurements that aim to measure different theory of mind abilities, from understanding false beliefs to interpreting indirect requests and recognizing irony and faux pas. We tested two families of...

[55] [Representation in large language models](#)

Cameron C. Yetman. ArXiv, 2025. 2 citations.

0% Topic Match

Abstract: The extraordinary success of recent Large Language Models (LLMs) on a diverse array of tasks has led to an explosion of scientific and philosophical theorizing aimed at explaining how they do what they do. Unfortunately, disagreement over fundamental theoretical issues has led to stalemate, with entrenched camps of LLM optimists and pessimists often committed to very different views of how these systems work. Overcoming stalemate requires agreement on fundamental questions, and the goal of this paper is to address one such question, namely: is LLM behavior driven partly by representation-based information processing of the sort implicated in biological cognition, or...

[56] [Large Language Models: The Need for Nuance in Current Debates and a Pragmatic Perspective on Understanding](#)

Bram van Dijk, ..., and Max J. van Duijn. ArXiv, 2023. 23 citations.

0% Topic Match

Abstract: Current Large Language Models (LLMs) are unparalleled in their ability to generate grammatically correct, fluent text. LLMs are appearing rapidly, and debates on LLM capacities have taken off, but reflection is lagging behind. Thus, in this position paper, we first zoom in on the debate and critically assess three points recurring in critiques of LLM capacities: i) that LLMs only parrot different statistical patterns in the training data; ii) that LLMs master formal but not functional language competence; and iii) that language learning in LLMs cannot inform human language learning. Drawing on empirical and theoretical arguments, we show that these points...

[57] [How FaR Are Large Language Models From Agents with Theory-of-Mind?](#)

Pei Zhou, ..., and Manaal Faruqi. ArXiv, 2023. 62 citations.

0% Topic Match

Abstract: "Thinking is for Doing." Humans can infer other people's mental states from observations--an ability called Theory-of-Mind (ToM)--and subsequently act pragmatically on those inferences. Existing question answering benchmarks such as ToMi ask models questions to make inferences about beliefs of characters in a story, but do not test whether models can then use these inferences to guide their actions. We propose a new evaluation paradigm for large language models (LLMs): Thinking for Doing (T4D), which requires models to connect inferences about others' mental states to actions in social scenarios. Experiments on T4D demonstrate that LLMs such as GPT-4 and PaLM 2 seemingly...

[58] [A Philosophical Introduction to Language Models - Part II: The Way Forward](#)

Raphael Milliere and Cameron Buckner. ArXiv, 2024. 21 citations.

0% Topic Match

Abstract: In this paper, the second of two companion pieces, we explore novel philosophical questions raised by recent progress in large language models (LLMs) that go beyond the classical debates covered in the first part. We focus particularly on issues related to interpretability, examining evidence from causal intervention methods about the nature of LLMs' internal representations and computations. We also discuss the implications of multimodal and modular extensions of LLMs, recent debates about whether such systems may meet minimal criteria for consciousness, and concerns about secrecy and reproducibility in LLM research. Finally, we discuss whether LLM-like systems may be relevant to...

[59] [Socially intelligent machines that learn from humans and help humans learn](#)

H. Gweon, ..., and Been Kim. Philosophical Transactions of the Royal Society A, 2023. 25 citations.

0% Topic Match

Abstract: A hallmark of human intelligence is the ability to understand and influence other minds. Humans engage in inferential social learning (ISL) by using commonsense psychology to learn from others and help others learn. Recent advances in artificial intelligence (AI) are raising new questions about the feasibility of human-machine interactions that support such powerful modes of social learning. Here, we envision what it means to develop socially intelligent machines that can learn, teach, and communicate in ways that are characteristic of ISL. Rather than machines that simply predict human behaviours or recapitulate superficial aspects of human sociality (e.g. smiling, imitating), we...

[60] [LLMs achieve adult human performance on higher-order theory of mind tasks](#)

Winnie Street, ..., and Robin I. M. Dunbar. ArXiv, 2024. 48 citations.

0% Topic Match

Abstract: This paper examines the extent to which large language models (LLMs) have developed higher-order theory of mind (ToM); the human ability to reason about multiple mental and emotional states in a recursive manner (e.g. I think that you believe that she knows). This paper builds on prior work by introducing a handwritten test suite -- Multi-Order Theory of Mind Q&A -- and using it to compare the performance of five LLMs to a newly gathered adult human benchmark. We find that GPT-4 and Flan-PaLM reach adult-level and near adult-level performance on ToM tasks overall, and that GPT-4 exceeds adult performance...

[61] [LLM Theory of Mind and Alignment: Opportunities and Risks](#)

Winnie Street. ArXiv, 2024. 12 citations.

0% Topic Match

Abstract: Large language models (LLMs) are transforming human-computer interaction and conceptions of artificial intelligence (AI) with their impressive capacities for conversing and reasoning in natural language. There is growing interest in whether LLMs have theory of mind (ToM); the ability to reason about the mental and emotional states of others that is core to human social intelligence. As LLMs are integrated into the fabric of our personal, professional and social lives and given greater agency to make decisions with real-world consequences, there is a critical need to understand how they can be aligned with human values. ToM seems to be a...

[62] [What if This Modified That? Syntactic Interventions with Counterfactual Embeddings](#)

Mycal Tucker, ..., and R. Levy. Unknown journal, 2021. 45 citations.

0% Topic Match

Abstract: Neural language models exhibit impressive performance on a variety of tasks, but their internal reasoning may be difficult to understand. Prior art aims to uncover meaningful properties within model representations via probes, but it is unclear how faithfully such probes portray information that the models actually use. To overcome such limitations, we propose a technique, inspired by causal analysis, for generating counterfactual embeddings within models. In experiments testing our technique, we produce evidence that suggests some BERT-based models use a tree-distance-like representation of syntax in downstream prediction tasks.

[63] [A Mechanistic Interpretation of Arithmetic Reasoning in Language Models using Causal Mediation Analysis](#)

Alessandro Stolfo, ..., and Mrinmaya Sachan. Unknown journal, 2023. 61 citations.

0% Topic Match

Abstract: Mathematical reasoning in large language models (LMs) has garnered significant attention in recent work, but there is a limited understanding of how these models process and store information related to arithmetic tasks within their architecture. In order to improve our understanding of this aspect of language models, we present a mechanistic interpretation of Transformer-based LMs on arithmetic questions using a causal mediation analysis framework. By intervening on the activations of specific model components and measuring the resulting changes in predicted probabilities, we identify the subset of parameters responsible for specific predictions. This provides insights into how information related to arithmetic...

[64] [Understanding Social Reasoning in Language Models with Language Models](#)

Kanishk Gandhi, ..., and Noah D. Goodman. ArXiv, 2023. 144 citations.

0% Topic Match

Abstract: As Large Language Models (LLMs) become increasingly integrated into our everyday lives, understanding their ability to comprehend human mental states becomes critical for ensuring effective interactions. However, despite the recent attempts to assess the Theory-of-Mind (ToM) reasoning capabilities of LLMs, the degree to which these models can align with human ToM remains a nuanced topic of exploration. This is primarily due to two distinct challenges: (1) the presence of inconsistent results from previous evaluations, and (2) concerns surrounding the validity of existing evaluation methodologies. To address these challenges, we present a novel framework for procedurally generating evaluations with LLMs by...

[65] [Large language models and linguistic intentionality](#)

J. Grindrod. ArXiv, 2024. 12 citations.

0% Topic Match

Abstract: Do large language models like Chat-GPT or Claude meaningfully use the words they produce? Or are they merely clever prediction machines, simulating language use by producing statistically plausible text? There have already been some initial attempts to answer this question by showing that these models meet the criteria for entering meaningful states according to metasegmental theories of mental content. In this paper, I will argue for a different approach—that we should instead consider whether language models meet the criteria given by our best metasegmental theories of linguistic content. In that vein, I will illustrate how this can be done by...

[66] [Theory of Mind Abilities of Large Language Models in Human-Robot Interaction: An Illusion?](#)

Mudit Verma, ..., and Subbarao Kambhampati. Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction, 2024. 25 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) have shown exceptional generative abilities in various natural language and generation tasks. However, possible anthropomorphization and leniency towards failure cases have propelled discussions on emergent abilities of LLMs especially on Theory of Mind (ToM) abilities in Large Language Models. While several false-belief tests exist to verify the ability to infer and maintain mental models of another entity, we study a special application of ToM abilities that has higher stakes and possibly irreversible consequences : Human Robot Interaction. In this work, we explore the task of Perceived Behavior Recognition, where a robot employs an LLM to assess...

[67] [Probing Language Models on Their Knowledge Source](#)

Zineddine Tighidet, ..., and P. Gallinari. ArXiv, 2024. 5 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) often encounter conflicts between their learned, internal (parametric knowledge, PK) and external knowledge provided during inference (contextual knowledge, CK). Understanding how LLMs models prioritize one knowledge source over the other remains a challenge. In this paper, we propose a novel probing framework to explore the mechanisms governing the selection between PK and CK in LLMs. Using controlled prompts designed to contradict the model's PK, we demonstrate that specific model activations are indicative of the knowledge source employed. We evaluate this framework on various LLMs of different sizes and demonstrate that mid-layer activations, particularly those related to...

[68] [Theory of Mind May Have Spontaneously Emerged in Large Language Models](#)

Michal Kosinski. ArXiv, 2023. 215 citations.

0% Topic Match

No summary or abstract available

[69] [Language Models are Few-Shot Learners](#)

Tom B. Brown, ..., and Dario Amodei. ArXiv, 2020. 48031 citations.

0% Topic Match

Abstract: Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions - something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically,...

[70] [The Remarkable Robustness of LLMs: Stages of Inference?](#)

Vedang Lad, ..., and Max Tegmark. ArXiv, 2024. 67 citations.

0% Topic Match

Abstract: We investigate the robustness of Large Language Models (LLMs) to structural interventions by deleting and swapping adjacent layers during inference. Surprisingly, models retain 72-95% of their original top-1 prediction accuracy without any fine-tuning. We find that performance degradation is not uniform across layers: interventions to the early and final layers cause the most degradation, while the model is remarkably robust to dropping middle layers. This pattern of localized sensitivity motivates our hypothesis of four stages of inference, observed across diverse model families and sizes: (1) detokenization, where local context is integrated to lift raw token embeddings into higher-level representations; (2)...

[71] [Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data](#)

Emily M. Bender and Alexander Koller. Unknown journal, 2020. 1025 citations.

0% Topic Match

Abstract: The success of the large neural language models on many NLP tasks is exciting. However, we find that these successes sometimes lead to hype in which these models are being described as "understanding" language or capturing "meaning". In this position paper, we argue that a system trained only on form has

a priori no way to learn meaning. In keeping with the ACL 2020 theme of "Taking Stock of Where We've Been and Where We're Going", we argue that a clear understanding of the distinction between form and meaning will help guide the field towards better science around natural language...

[72] [The Reasoning-Memorization Interplay in Language Models Is Mediated by a Single Direction](#)

Yihuai Hong, ..., and Zhijing Jin. ArXiv, 2025. 9 citations.

0% Topic Match

Abstract: Large language models (LLMs) excel on a variety of reasoning benchmarks, but previous studies suggest they sometimes struggle to generalize to unseen questions, potentially due to over-reliance on memorized training examples. However, the precise conditions under which LLMs switch between reasoning and memorization during text generation remain unclear. In this work, we provide a mechanistic understanding of LLMs' reasoning-memorization dynamics by identifying a set of linear features in the model's residual stream that govern the balance between genuine reasoning and memory recall. These features not only distinguish reasoning tasks from memory-intensive ones but can also be manipulated to causally influence...

[73] [Large Language Models Fail on Trivial Alterations to Theory-of-Mind Tasks](#)

T. Ullman. ArXiv, 2023. 273 citations.

0% Topic Match

Abstract: Intuitive psychology is a pillar of common-sense reasoning. The replication of this reasoning in machine intelligence is an important stepping-stone on the way to human-like artificial intelligence. Several recent tasks and benchmarks for examining this reasoning in Large-Language Models have focused in particular on belief attribution in Theory-of-Mind tasks. These tasks have shown both successes and failures. We consider in particular a recent purported success case, and show that small variations that maintain the principles of ToM turn the results on their head. We argue that in general, the zero-hypothesis for model evaluation in intuitive psychology should be skeptical, and...

[74] [False Sense of Security: Why Probing-based Malicious Input Detection Fails to Generalize](#)

Cheng Wang, ..., and Muhao Chen. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) can comply with harmful instructions, raising serious safety concerns despite their impressive capabilities. Recent work has leveraged probing-based approaches to study the separability of malicious and benign inputs in LLMs' internal representations, and researchers have proposed using such probing methods for safety detection. We systematically re-examine this paradigm. Motivated by poor out-of-distribution performance, we hypothesize that probes learn superficial patterns rather than semantic harmfulness. Through controlled experiments, we confirm this hypothesis and identify the specific patterns learned: instructional patterns and trigger words. Our investigation follows a systematic approach, progressing from demonstrating comparable performance of simple n-gram methods,...

[75] [Large Language Models Encode Semantics in Low-Dimensional Linear Subspaces](#)

Baturay Sa lam, ..., and Amin Karbasi ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Understanding the latent space geometry of large language models (LLMs) is key to interpreting their behavior and improving alignment. However, it remains unclear to what extent LLMs internally organize representations related to semantic understanding. To explore this, we conduct a large-scale empirical study of hidden representations in 11 autoregressive models across 6 scientific topics. We find that high-level semantic information consistently resides in low-dimensional subspaces that form linearly separable representations across domains. This separability becomes more pronounced in deeper layers and under prompts that elicit structured reasoning or alignment behavior\$\\\$even when surface content remains unchanged. These findings support geometry-aware tools...

[76] [Are explicit belief representations necessary? A comparison between Large Language Models and Bayesian probabilistic models](#)

Dingyi Pan and Benjamin K. Bergen. Unknown journal, 2025. 0 citations.

0% Topic Match

Abstract: Large language models (LLMs) have exhibited certain indirect pragmatic capabilities, including interpreting indirect requests and non-literal meanings. Yet, it is unclear whether the success of LLMs on pragmatic tasks generalizes to phenomena that directly probe inferences about the beliefs of others. Indeed, LLMs' performance on Theory of Mind (ToM) tasks is mixed. To date, the most successful computationally explicit approach to making inferences about others' beliefs is the Rational Speech Act (RSA) framework, a Bayesian probabilistic model that encodes explicit representations of beliefs. In the present study, we ask whether LLMs outperform RSA in predicting human belief inferences, even though...

[77] [Evaluating Large Language Models Trained on Code](#)

Mark Chen, ..., and Wojciech Zaremba. ArXiv, 2021. 6433 citations.

0% Topic Match

Abstract: We introduce Codex, a GPT language model fine-tuned on publicly available code from GitHub, and study its Python code-writing capabilities. A distinct production version of Codex powers GitHub Copilot. On HumanEval, a new evaluation set we release to measure functional correctness for synthesizing programs from docstrings, our model solves 28.8% of the problems, while GPT-3 solves 0% and GPT-J solves 11.4%. Furthermore, we find that repeated sampling from the model is a surprisingly effective strategy for producing working solutions to difficult prompts. Using this method, we solve 70.2% of our problems with 100 samples per problem. Careful investigation of our...

[78] [Is German secretly a Slavic language? What BERT probing can tell us about language groups](#)

Aleksandra Mysiak and J. Cyranka. Proceedings of the 9th Workshop on Slavic Natural Language Processing 2023 (SlavicNLP 2023), 2023. 2 citations.

0% Topic Match

Abstract: In the light of recent developments in NLP, the problem of understanding and interpreting large language models has gained a lot of urgency. Methods developed to study this area are subject to considerable scrutiny. In this work, we take a closer look at one such method, the structural probe introduced by Hewitt and Manning (2019). We run a series of experiments involving multiple languages, focusing principally on the group of Slavic languages. We show that probing results can be seen as a reflection of linguistic classification, and conclude that multilingual BERT learns facts about languages and their groups.

[79] [Do We Know What LLMs Don't Know? A Study of Consistency in Knowledge Probing](#)

Raoyuan Zhao, ..., and Hinrich Schutze. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: The reliability of large language models (LLMs) is greatly compromised by their tendency to hallucinate, underscoring the need for precise identification of knowledge gaps within LLMs. Various methods for probing such gaps exist, ranging from calibration-based to prompting-based methods. To evaluate these probing methods, in this paper, we propose a new process based on using input variations and quantitative metrics. Through this, we expose two dimensions of inconsistency in knowledge gap probing. (1) Intra-method inconsistency: Minimal non-semantic perturbations in prompts lead to considerable variance in detected knowledge gaps within the same probing method; e.g., the simple variation of shuffling answer...

[80] [Clever Hans or Neural Theory of Mind? Stress Testing Social Reasoning in Large Language Models](#)

Natalie Shapira, ..., and Vered Shwartz. Unknown journal, 2023. 154 citations.

0% Topic Match

Abstract: The escalating debate on AI's capabilities warrants developing reliable metrics to assess machine "intelligence." Recently, many anecdotal examples were used to suggest that newer Large Language Models (LLMs) like ChatGPT and GPT-4 exhibit Neural Theory-of-Mind (N-ToM); however, prior work reached conflicting conclusions regarding those abilities. We investigate the extent of LLMs' N-ToM through an extensive evaluation of 6 tasks and find that while LLMs exhibit certain N-ToM abilities, this behavior is far from being robust. We further examine the factors impacting performance on N-ToM tasks and discover that LLMs struggle with adversarial examples, indicating reliance on shallow heuristics rather than...

[81] [Do Multimodal Large Language Models and Humans Ground Language Similarly?](#)

Cameron R. Jones, ..., and Sean Trott. Computational Linguistics, 2024. 5 citations.

0% Topic Match

Abstract: Abstract Large Language Models (LLMs) have been criticized for failing to connect linguistic meaning to the world—for failing to solve the "symbol grounding problem." Multimodal Large Language Models (MLLMs) offer a potential solution to this challenge by combining linguistic representations and processing with other modalities. However, much is still unknown about exactly how and to what degree MLLMs integrate their distinct modalities—and whether the way they do so mirrors

the mechanisms believed to underpin grounding in humans. In humans, it has been hypothesized that linguistic meaning is grounded through "embodied simulation," the activation of sensorimotor and affective representations reflecting described...

[82] [Probing Syntax in Large Language Models: Successes and Remaining Challenges](#)

Pablo J. Diego-Sim'on, ..., and Yair Lakretz. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: The syntactic structures of sentences can be readily read-out from the activations of large language models (LLMs). However, the "structural probes" that have been developed to reveal this phenomenon are typically evaluated on an indiscriminate set of sentences. Consequently, it remains unclear whether structural and/or statistical factors systematically affect these syntactic representations. To address this issue, we conduct an in-depth analysis of structural probes on three controlled benchmarks. Our results are three-fold. First, structural probes are biased by a superficial property: the closer two words are in a sentence, the more likely structural probes will consider them as syntactically linked. Second,...

[83] [What is consciousness, and could machines have it?](#)

S. Dehaene, ..., and S. Kouider. Science, 2017. 563 citations.

0% Topic Match

Abstract: The controversial question of whether machines may ever be conscious must be based on a careful consideration of how consciousness arises in the only physical system that undoubtedly possesses it: the human brain. We suggest that the word "consciousness" conflates two different types of information-processing computations in the brain: the selection of information for global broadcasting, thus making it flexibly available for computation and report (C1, consciousness in the first sense), and the self-monitoring of those computations, leading to a subjective sense of certainty or error (C2, consciousness in the second sense). We argue that despite their recent successes, current...

[84] [Logical forms complement probability in understanding language model \(and human\) performance](#)

Yixuan Wang and Freda Shi. ArXiv, 2025. 2 citations.

0% Topic Match

Abstract: With the increasing interest in using large language models (LLMs) for planning in natural language, understanding their behaviors becomes an important research question. This work conducts a systematic investigation of LLMs' ability to perform logical reasoning in natural language. We introduce a controlled dataset of hypothetical and disjunctive syllogisms in propositional and modal logic and use it as the testbed for understanding LLM performance. Our results lead to novel insights in predicting LLM behaviors: in addition to the probability of input (Gonen et al., 2023; McCoy et al., 2024), logical forms should be considered as important factors. In addition, we...

[85] [Predicting the Performance of Black-box LLMs through Self-Queries](#)

Dylan Sam, ..., and J. Kolter. ArXiv, 2025. 7 citations.

0% Topic Match

Abstract: As large language models (LLMs) are increasingly relied on in AI systems, predicting when they make mistakes is crucial. While a great deal of work in the field uses internal representations to interpret model behavior, these representations are inaccessible when given solely black-box access through an API. In this paper, we extract features of LLMs in a black-box manner by using follow-up prompts and taking the probabilities of different responses as representations to train reliable predictors of model behavior. We demonstrate that training a linear model on these low-dimensional representations produces reliable and generalizable predictors of model performance at the...

[86] [Probing Across Time: What Does RoBERTa Know and When?](#)

Leo Z. Liu, ..., and Noah A. Smith. Unknown journal, 2021. 89 citations.

0% Topic Match

Abstract: Models of language trained on very large corpora have been demonstrated useful for NLP. As fixed artifacts, they have become the object of intense study, with many researchers "probing" the extent to which linguistic abstractions, factual and commonsense knowledge, and reasoning abilities they acquire and readily demonstrate. Building on this line of work, we consider a new question: for types of knowledge a language model learns, when during (pre)training are they acquired? We plot probing performance across iterations, using RoBERTa as a case study. Among our findings: linguistic knowledge is acquired fast, stably, and robustly across domains. Facts and commonsense are slower...

[87] [Consciousness in Artificial Intelligence: Insights from the Science of Consciousness](#)

Patrick Butlin, ..., and R. VanRullen. ArXiv, 2023. 129 citations.

0% Topic Match

Abstract: Whether current or near-term AI systems could be conscious is a topic of scientific interest and increasing public concern. This report argues for, and exemplifies, a rigorous and empirically grounded approach to AI consciousness: assessing existing AI systems in detail, in light of our best-supported neuroscientific theories of consciousness. We survey several prominent scientific theories of consciousness, including recurrent processing theory, global workspace theory, higher-order theories, predictive processing, and attention schema theory. From these theories we derive "indicator properties" of consciousness, elucidated in computational terms that allow us to assess AI systems for these properties. We use these indicator properties to assess...

[88] [FANToM: A Benchmark for Stress-testing Machine Theory of Mind in Interactions](#)

Hyunwoo Kim, ..., and Maarten Sap. ArXiv, 2023. 111 citations.

0% Topic Match

Abstract: Theory of mind (ToM) evaluations currently focus on testing models using passive narratives that inherently lack interactivity. We introduce FANToM, a new benchmark designed to stress-test ToM within information-asymmetric conversational contexts via question answering. Our benchmark draws upon important theoretical requisites from psychology and necessary empirical considerations when evaluating large language models (LLMs). In particular, we formulate multiple types of questions that demand the same underlying reasoning to identify illusory or false sense of ToM capabilities in LLMs. We show that FANToM is challenging for state-of-the-art LLMs, which perform significantly worse than humans even with chain-of-thought reasoning or fine-tuning.

[89] [EPITOME: Experimental Protocol Inventory for Theory Of Mind Evaluation](#)

Cameron R. Jones, ..., and Benjamin Bergen. Journal Not Provided, Unknown year. 10 citations.

0% Topic Match

Abstract: We address a growing debate about the extent to which large language models (LLMs) produce behavior consistent with Theory of Mind (ToM) in humans. We present EPITOME: a battery of six experiments that tap diverse ToM capacities, including belief attribution, emotional inference, and pragmatic reasoning. We compare performance of five LLMs to a baseline of responses from human comprehenders. Results are mixed. LLMs display considerable sensitivity to mental states and match human performance in several tasks. Yet, they commit systematic errors in others, especially those requiring pragmatic reasoning on the basis of mental state information. Such uneven performance indicates that...

[90] [The Curious Case of Representational Alignment: Unravelling Visio-Linguistic Tasks in Emergent Communication](#)

Tom Kouwenhoven, ..., and Tessa Verhoef. ArXiv, 2024. 8 citations.

0% Topic Match

Abstract: Natural language has the universal properties of being compositional and grounded in reality. The emergence of linguistic properties is often investigated through simulations of emergent communication in referential games. However, these experiments have yielded mixed results compared to similar experiments addressing linguistic properties of human language. Here we address representational alignment as a potential contributing factor to these results. Specifically, we assess the representational alignment between agent image representations and between agent representations and input images. Doing so, we confirm that the emergent language does not appear to encode human-like conceptual visual features, since agent image representations drift away from...

[91] [A Latent-Variable Model for Intrinsic Probing](#)

Karolina Stańczak, ..., and Isabelle Augenstein. ArXiv, 2022. 4 citations.

0% Topic Match

Abstract: The success of pre-trained contextualized representations has prompted researchers to analyze them for the presence of linguistic information. Indeed, it is natural to assume that these pre-trained representations do encode some level of linguistic knowledge as they have brought about large empirical improvements on a wide variety of NLP tasks, which suggests they are learning true linguistic generalization. In this work, we focus on intrinsic probing, an analysis technique where the goal is not only to identify whether a representation encodes a linguistic attribute but also to pinpoint where this attribute is encoded. We propose a novel latent-variable formulation for...

[92] [Moral consideration for AI systems by 2030](#)

Jeff Sebo and Robert Long. AI Ethics, 2023. 24 citations.

0% Topic Match

Abstract: This paper makes a simple case for extending moral consideration to some AI systems by 2030. It involves a normative premise and a descriptive premise. The normative premise is that humans have a duty to extend moral consideration to beings that have a non-negligible chance, given the evidence, of being conscious. The descriptive premise is that some AI systems do in fact have a non-negligible chance, given the evidence, of being conscious by 2030. The upshot is that humans have a duty to extend moral consideration to some AI systems by 2030. And if we have a duty to do...

[93] [Towards Interpretable Sequence Continuation: Analyzing Shared Circuits in Large Language Models](#)

Michael Lan, ..., and Fazl Barez. Unknown journal, 2023. 7 citations.

0% Topic Match

Abstract: While transformer models exhibit strong capabilities on linguistic tasks, their complex architectures make them difficult to interpret. Recent work has aimed to reverse engineer transformer models into human-readable representations called circuits that implement algorithmic functions. We extend this research by analyzing and comparing circuits for similar sequence continuation tasks, which include increasing sequences of Arabic numerals, number words, and months. By applying circuit interpretability analysis, we identify a key sub-circuit in both GPT-2 Small and Llama-2-7B responsible for detecting sequence members and for predicting the next member in a sequence. Our analysis reveals that semantically related sequences rely on shared...

[94] [MMToM-QA: Multimodal Theory of Mind Question Answering](#)

Chuanyang Jin, ..., and Tianmin Shu. ArXiv, 2024. 61 citations.

0% Topic Match

Abstract: Theory of Mind (ToM), the ability to understand people's mental states, is an essential ingredient for developing machines with human-level social intelligence. Recent machine learning models, particularly large language models, seem to show some aspects of ToM understanding. However, existing ToM benchmarks use unimodal datasets - either video or text. Human ToM, on the other hand, is more than video or text understanding. People can flexibly reason about another person's mind based on conceptual representations (e.g., goals, beliefs, plans) extracted from any available data. To address this, we introduce a multimodal Theory of Mind question answering (MMToM-QA) benchmark. MMToM-QA comprehensively...

[95] [Artificial Intelligence and the Illusion of Understanding: A Systematic Review of Theory of Mind and Large Language Models](#)

Antonella Marchetti, ..., and D. Massaro. Cyberpsychology, behavior and social networking, 2025. 1 citations.

0% Topic Match

Abstract: The development of Large Language Models (LLMs) has sparked significant debate regarding their capacity for Theory of Mind (ToM)-the ability to attribute mental states to oneself and others. This systematic review examines the extent to which LLMs exhibit Artificial ToM (AToM) by evaluating their performance on ToM tasks and comparing it with human responses. While LLMs, particularly GPT-4, perform well on first-order false belief tasks, they struggle with more complex reasoning, such as second-order beliefs and recursive inferences, where humans consistently outperform them. Moreover, the review underscores the variability in ToM assessments, as many studies adapt classical tasks for LLMs,...

[96] [Explore Theory of Mind: Program-guided adversarial data generation for theory of mind reasoning](#)

Melanie Sclar, ..., and Asli Celikyilmaz. ArXiv, 2024. 16 citations.

0% Topic Match

Abstract: Do large language models (LLMs) have theory of mind? A plethora of papers and benchmarks have been introduced to evaluate if current models have been able to develop this key ability of social intelligence. However, all rely on limited datasets with simple patterns that can potentially lead to problematic blind spots in evaluation and an overestimation of model capabilities. We introduce ExploreToM, the first framework to allow large-scale generation of diverse and challenging theory of mind data for robust training and evaluation. Our approach leverages an A* search over a custom domain-specific language to produce complex story structures and novel,...

[97] [Relational composition during attribute retrieval in GPT is not purely linear](#)

Michael B. McCoy and Anna Leshinskaya. Journal Not Provided, Unknown year. 0 citations.

0% Topic Match

Abstract: The longstanding question of how neural networks could implement relational composition has been buoyed by recent success showing relational abstraction in transformer-based large language models (LLMs). We address recent findings showing some, but imperfect, generalizability in linear composition during knowledge retrieval of attributive triplets such as has-color (banana, yellow) in GPT-J [Hernandez, E. et al, (2024). Linearity of relation decoding in transformer language models (arXiv:2308.09124)]. We report that limitations to relational generalization are explainable by two systematic factors. First, relational combinations that are more accurately retrieved generalize better than uncertain or inaccurate ones. Second, relational generalization scales with the semantic...

[98] [Sensitivity Meets Sparsity: The Impact of Extremely Sparse Parameter Patterns on Theory-of-Mind of Large Language Models](#)

Yuheng Wu, ..., and Denghui Zhang. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: This paper investigates the emergence of Theory-of-Mind (ToM) capabilities in large language models (LLMs) from a mechanistic perspective, focusing on the role of extremely sparse parameter patterns. We introduce a novel method to identify ToM-sensitive parameters and reveal that perturbing as little as 0.001% of these parameters significantly degrades ToM performance while also impairing contextual localization and language understanding. To understand this effect, we analyze their interaction with core architectural components of LLMs. Our findings demonstrate that these sensitive parameters are closely linked to the positional encoding module, particularly in models using Rotary Position Embedding (RoPE), where perturbations disrupt dominant-frequency...

[99] [Tests of Animal Consciousness are Tests of Machine Consciousness](#)

L. Dung. Erkenntnis, 2023. 13 citations.

0% Topic Match

Abstract: If a machine attains consciousness, how could we find out? In this paper, I make three related claims regarding positive tests of machine consciousness. All three claims center on the idea that an AI can be constructed "ad hoc", that is, with the purpose of satisfying a particular test of consciousness while clearly not being conscious. First, a proposed test of machine consciousness can be legitimate, even if AI can be constructed ad hoc specifically to pass this test. This is underscored by the observation that many, if not all, putative tests of machine consciousness can be passed by non-conscious...

[100] [Circuit Compositions: Exploring Modular Structures in Transformer-Based Language Models](#)

Philipp Mondorf, ..., and Barbara Plank. ArXiv, 2024. 2 citations.

0% Topic Match

Abstract: A fundamental question in interpretability research is to what extent neural networks, particularly language models, implement reusable functions through subnetworks that can be composed to perform more complex tasks. Recent advances in mechanistic interpretability have made progress in identifying circuits , which represent the minimal computational subgraphs responsible for a model's behavior on specific tasks. However, most studies focus on identifying circuits for individual tasks without investigating how functionally similar circuits relate to each other. To address this gap, we study the modularity of neural networks by analyzing circuits for highly compositional subtasks within a transformer-based language model. Specifically, given a...

[101] [CausaLM: Causal Model Explanation Through Counterfactual Language Models](#)

Amir Feder, ..., and Roi Reichart. Computational Linguistics, 2020. 168 citations.

0% Topic Match

Abstract: Abstract Understanding predictions made by deep neural networks is notoriously difficult, but also crucial to their dissemination. As all machine learning-based methods, they are as good as their training data, and can also capture unwanted biases. While there are tools that can help understand whether such biases exist, they do not distinguish between correlation and causation, and might be ill-suited for text-based models and for reasoning about high-level language concepts. A key problem of estimating the causal effect of a concept of interest on a given model is that this estimation requires the generation of counterfactual examples, which is challenging...

[102] [Dissociable Neural Mechanisms for Human Inference Processing Predicted by Static and Contextual Language Models](#)

Takahisa Uchida, ..., and P. Dominey. Neurobiology of Language, 2022. 3 citations.

0% Topic Match

Abstract: Abstract Language models (LMs) continue to reveal non-trivial relations to human language performance and the underlying neurophysiology. Recent research has characterized how word embeddings from an LM can be used to generate integrated discourse representations in order to perform inference on events. The current research investigates how such event knowledge may be coded in distinct manners in different classes of LMs and how this maps onto different forms of human inference processing. To do so, we investigate inference on events using two well-documented human experimental protocols from Metusalem et al. (2012) and McKoon and Ratcliff (1986), compared with two protocols...

[103] [Digital suffering: why it's a problem and how to prevent it](#)

Bradford Saad and A. Bradley. Inquiry, 2022. 8 citations.

0% Topic Match

Abstract: As ever more advanced digital systems are created, it becomes increasingly likely that some of these systems will be digital minds, i.e. digital subjects of experience. With digital minds comes the risk of digital suffering. The problem of digital suffering is that of mitigating this risk. We argue that the problem of digital suffering is a high stakes moral problem and that formidable epistemic obstacles stand in the way of solving it. We then propose a strategy for solving it: Access Monitor Prevent (AMP). AMP uses a 'dancing qualia' argument to...

[104] [Beyond Pattern Recognition: Probing Mental Representations of LMs](#)

Moritz Miller and Kumar Shridhar. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Language Models (LMs) have demonstrated impressive capabilities in solving complex reasoning tasks, particularly when prompted to generate intermediate explanations. However, it remains an open question whether these intermediate reasoning traces represent a dynamic, evolving thought process or merely reflect sophisticated pattern recognition acquired during large scale pre training. Drawing inspiration from human cognition, where reasoning unfolds incrementally as new information is assimilated and internal models are continuously updated, we propose to delve deeper into the mental model of various LMs. We propose a new way to assess the mental modeling of LMs, where they are provided with problem details gradually...

[105] [Evaluating large language models in theory of mind tasks](#)

Michal Kosinski. Proceedings of the National Academy of Sciences of the United States of America, 2023. 196 citations.

0% Topic Match

Abstract: Significance Humans automatically and effortlessly track others' unobservable mental states, such as their knowledge, intentions, beliefs, and desires. This ability—typically called "theory of mind" (ToM)—is fundamental to human social interactions, communication, empathy, consciousness, moral judgment, and religious beliefs. Our results show that recent large language models (LLMs) can solve false-belief tasks, typically used to evaluate ToM in humans. Regardless of how we interpret these outcomes, they signify the advent of more powerful and socially skilled AI—with profound positive and negative implications.

[106] [Error Detection and Representational Mechanisms](#)

K. Bielecka and M. MiBkowski. Unknown journal, 2020. 6 citations.

0% Topic Match

Abstract: This chapter defends a mechanistic and teleosemantic view of naturalized intentionality that underlies the role of error detection via coherence checking. Representational mechanisms serve the biological functions of representing, which are related to the semantic value of representation: its truth or falsity, its being vacuous or satisfied, or its accuracy. If representational mechanisms contain (or interact with) error-detection mechanisms, the semantic value of representation is causally relevant. As long as semantic value is causally relevant in cognitive explanations, the content of representation is arguably causally relevant, which vindicates the notion of mental representation in contemporary scientific research. Error detection is...

[107] [How to deal with risks of AI suffering](#)

L. Dung. Inquiry, 2023. 7 citations.

0% Topic Match

No summary or abstract available

[108] [ECM: A Unified Electronic Circuit Model for Explaining the Emergence of In-Context Learning and Chain-of-Thought in Large Language Model](#)

Qiguang Chen, ..., and Ting Liu. ArXiv, 2025. 4 citations.

0% Topic Match

Abstract: Recent advancements in large language models (LLMs) have led to significant successes across various applications, where the most noticeable is to a series of emerging capabilities, particularly in the areas of In-Context Learning (ICL) and Chain-of-Thought (CoT). To better understand and control model performance, many studies have begun investigating the underlying causes of these phenomena and their impact on task outcomes. However, existing explanatory frameworks predominantly focus on isolating and explaining ICL and CoT independently, leading to an incomplete understanding of their combined influence on model performance. To address this gap, we propose the Electronic Circuit Model (ECM), which provides...

[109] [Language Models Can Predict Their Own Behavior](#)

Dhananjay Ashok and Jonathan May. ArXiv, 2025. 5 citations.

0% Topic Match

Abstract: The text produced by language models (LMs) can exhibit specific 'behaviors,' such as a failure to follow alignment training, that we hope to detect and react to during deployment. Identifying these behaviors can often only be done post facto, i.e., after the entire text of the output has been generated. We provide evidence that there are times when we can predict how an LM will behave early in computation, before even a single token is generated. We show that probes trained on the internal representation of input tokens alone can predict a wide range of eventual behaviors over the entire output...

[110] [Do Theory of Mind Benchmarks Need Explicit Human-like Reasoning in Language Models?](#)

Yi-Long Lu, ..., and Wei Wang. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: Theory of Mind (ToM), the ability to attribute mental states to others, is fundamental for human social intelligence and a critical capability for advanced Artificial Intelligence. Recent advancements in Large Language Models (LLMs) have shown promising performance on ToM benchmarks, raising the question: Do these benchmarks necessitate explicit human-like reasoning processes, or can models succeed through alternative strategies? We investigate this question empirically by applying Reinforcement Learning (RL) and Supervised Fine-Tuning (SFT) to LLMs of varying scales (0.5B to 7B parameters) and evaluating them across multiple ToM datasets. Our results reveal a scale-dependent impact of RL: while RL significantly improves...

[111] [Exploring Alternative Approaches to Language Modeling for Learning from Data and Knowledge](#)

Yuxin Zi, ..., and Amit P. Sheth. Unknown journal, 2024. 1 citations.

0% Topic Match

Abstract: Despite their extensive application in language understanding tasks, large language models (LLMs) still encounter challenges including hallucinations - occasional fabrication of information - and alignment issues - lack of associations with human-curated world models (e.g., intuitive physics or common-sense knowledge). Moreover, the black-box nature of LLMs presents significant obstacles in training them effectively to achieve desired behaviors. In particular, modifying the concept embedding spaces of LLMs can be highly intractable. This process involves analyzing the implicit impact of such adjustments on the myriad parameters within LLMs and the resulting inductive biases. We propose a novel architecture that wraps powerful function...

[112] [Sparks of Artificial General Intelligence: Early experiments with GPT-4](#)

Sébastien Bubeck, ..., and Yi Zhang. ArXiv, 2023. 3509 citations.

0% Topic Match

Abstract: Artificial intelligence (AI) researchers have been developing and refining large language models (LLMs) that exhibit remarkable capabilities across a variety of domains and tasks, challenging our understanding of learning and cognition. The latest model developed by OpenAI, GPT-4, was trained using an unprecedented scale of compute and data. In this paper, we report on our investigation of an early version of GPT-4, when it was still in active development by OpenAI. We contend that (this early version of) GPT-4 is part of a new cohort of LLMs (along with ChatGPT and Google's PaLM for example) that exhibit more general intelligence...

[113] [General Intelligence: An Ecumenical Heuristic for Artificial Consciousness Research?](#)

Henry Shevlin. J. Artif. Intell. Conscious., 2020. 7 citations.

0% Topic Match

Abstract: The science of consciousness has made great strides in recent decades. However, the proliferation of competing theories makes it difficult to reach consensus about artificial consciousness. While for purely scientific purposes we might wish to adopt a 'wait and see' attitude, we may soon face practical and ethical questions about whether, for example, agents artificial systems are capable of suffering. Moreover, many of the methods used for assessing consciousness in humans and even non-human animals are not straightforwardly applicable to artificial systems. With these challenges in mind, I propose that we look for ecumenical heuristics for artificial consciousness to enable...

[114] [Disembodied Meaning? Generative AI and Understanding](#)

Jordi Vallverdú and Iván Redondo. Forum for Linguistic Studies, 2025. 1 citations.

0% Topic Match

Abstract: This study explores the cognitive and philosophical implications of Large Language Models (LLMs), focusing on their ability to generate meaning without embodiment. Grounded in the coherence-based semantics framework, the research challenges traditional views that emphasize the necessity of embodied cognition for meaningful language comprehension. Through a theoretical and comparative analysis, this paper examines the limitations of embodied cognition paradigms, such as the symbol grounding problem and critiques like Searle's Chinese Room, and evaluates the practical capabilities of LLMs. The methodology integrates philosophical inquiry with empirical evidence, including case studies on LLM performance in tasks such as medical licensing exams, multilingual...

[115] [Multi-Agent Language Models: Advancing Cooperation, Coordination, and Adaptation](#)

Arjun Vaithilingam Sudhakar. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Modern Large Language Models (LLMs) exhibit impressive zero-shot and few-shot generalization capabilities across complex natural language tasks, enabling their widespread use as virtual assistants for diverse applications such as translation and summarization. Despite being trained solely on large corpora of text without explicit supervision on author intent, LLMs appear to infer the underlying meaning of textual interactions. This raises a fundamental question: can LLMs model and reason about the intentions of others, i.e., do they possess a form of theory of mind? Understanding other's intentions is crucial for effective collaboration, which underpins human societal success and is essential for cooperative...

[116] [A Philosophical Introduction to Language Models - Part I: Continuity With Classic Debates](#)

Raphael Milliere and Cameron Buckner. ArXiv, 2024. 30 citations.

0% Topic Match

Abstract: Large language models like GPT-4 have achieved remarkable proficiency in a broad spectrum of language-based tasks, some of which are traditionally associated with hallmarks of human intelligence. This has prompted ongoing disagreements about the extent to which we can meaningfully ascribe any kind of linguistic or cognitive competence to language models. Such questions have deep philosophical roots, echoing longstanding debates about the status of artificial neural networks as cognitive models. This article -- the first part of two companion papers -- serves both as a primer on language models for philosophers, and as an opinionated survey of their significance in...

[117] [Implementing artificial consciousness](#)

Leonard Dung and Luke Kersten. Mind & Language, 2024. 1 citations.

0% Topic Match

Abstract: Implementationism maintains that conventional, silicon based artificial systems are not conscious because they fail to satisfy certain substantive constraints on computational implementation. In this article, we argue that several recently proposed substantive constraints are implausible, or at least are not well supported, insofar as they conflate intuitions about computational implementation generally and consciousness specifically. We argue instead that the mechanistic account of computation can explain several of the intuitions driving implementationism and non computationalism in a manner which is consistent with artificial consciousness. Our argument provides indirect support for computationalism about consciousness and the view that conventional artificial systems can be conscious.

[118] [Deconfounded Causality-aware Parameter-Efficient Fine-Tuning for Problem-Solving Improvement of LLMs](#)

Ruoyu Wang, ..., and Lina Yao. Unknown journal, 2024. 0 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) have demonstrated remarkable efficiency in tackling various tasks based on human instructions, but studies reveal that they often struggle with tasks requiring reasoning, such as math or physics. This limitation raises questions about whether LLMs truly comprehend embedded knowledge or merely learn to replicate the token distribution without a true understanding of the content. In this paper, we delve into this problem and aim to enhance the reasoning capabilities of LLMs. First, we investigate if the model has genuine reasoning capabilities by visualizing the text generation process at the attention and representation level. Then, we formulate...

[119] [LLMs for Relational Reasoning: How Far are We?](#)

Zhiming Li, ..., and Yang Liu. 2024 IEEE/ACM International Workshop on Large Language Models for Code (LLM4Code), 2024. 35 citations.

0% Topic Match

Abstract: Large language models (LLMs) have revolutionized many areas (e.g. natural language processing, software engineering, etc.) by achieving state-of-the-art performance on extensive downstream tasks. Aiming to achieve robust and general artificial intelligence, there has been a surge of interest in investigating the reasoning ability of the LLMs. Whereas the textual and numerical reasoning benchmarks adopted by previous works are rather shallow and simple, it is hard to conclude that the LLMs possess strong reasoning ability by merely achieving positive results on these benchmarks. Recent efforts have demonstrated that the LLMs are poor at solving sequential decision-making problems that require common-sense planning...

[120] [Mind Over Machine: Evaluating Theory of Mind Reasoning in LLMs and Humans](#)

Maryam Amirizani. Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, 2025. 1 citations.

0% Topic Match

Abstract: As Large Language Models (LLMs) become increasingly integrated into applications demanding human-like understanding-such as mental health support, education, and social robotics-their capacity to exhibit Theory of Mind (ToM) reasoning is essential. Although previous research has evaluated LLMs' capability in ToM tasks, a critical gap remains as few studies have systematically investigated how LLMs' ToM reasoning diverges from human reasoning and the extent of these differences. This study introduces a reinforcement learning-based framework designed to bridge this gap. This approach seeks to enhance LLM alignment with human ToM reasoning, effectively narrowing the differences in their reasoning processes. Finally, future research directions...

[121] [Making Natural Language Reasoning Explainable and Faithful](#)

Xinya Du. Unknown journal, 2024. 2 citations.

0% Topic Match

Abstract: Neural models, including large language models (LLMs), achieve superior performance on logical reasoning tasks such as question answering. To elicit reasoning capabilities from LLMs, recent works propose using the chain-of-thought (CoT) mechanism to generate both the reasoning chain and the answer, which enhances the model's capabilities in conducting reasoning. However, due to LLM's uninterpretable nature and the extreme flexibility of free-form explanations, several challenges remain: such as struggling with inaccurate reasoning, hallucinations, and not aligning with human preferences. In this talk, we will focus on (1) our design of leveraging structured information (that is grounded to the context), for the...

[122] [Assessing tests of animal consciousness](#)

L. Dung. Consciousness and Cognition, 2022. 14 citations.

0% Topic Match

No summary or abstract available

[123] [DeepThought: An Architecture for Autonomous Self-motivated Systems](#)

Arlindo L. Oliveira, ..., and Pedro U. Lima. ArXiv, 2023. 0 citations.

0% Topic Match

Abstract: The ability of large language models (LLMs) to engage in credible dialogues with humans, taking into account the training data and the context of the conversation, has raised discussions about their ability to exhibit intrinsic motivations, agency, or even some degree of consciousness. We argue that the internal architecture of LLMs and their finite and volatile state cannot support any of these properties. By combining insights from complementary learning systems, global neuronal workspace, and attention schema theories, we propose to integrate LLMs and other deep learning systems into an architecture for cognitive language agents able to exhibit properties akin to...

[124] [Meta-Models: An Architecture for Decoding LLM Behaviors Through Interpreted Embeddings and Natural Language](#)

Anthony Costarelli, ..., and Severin Field. ArXiv, 2024. 3 citations.

0% Topic Match

Abstract: As Large Language Models (LLMs) become increasingly integrated into our daily lives, the potential harms from deceptive behavior underlie the need for faithfully interpreting their decision-making. While traditional probing methods have shown some effectiveness, they remain best for narrowly scoped tasks while more comprehensive explanations are still necessary. To this end, we investigate meta-models-an architecture using a "meta-model" that takes activations from an "input-model" and answers natural language questions about the input-model's behaviors. We evaluate the meta-model's ability to generalize by training them on selected task types and assessing their out-of-distribution performance in deceptive scenarios. Our findings show that meta-models generalize well to out-of-distribution...

[125] [Misinforming LLMs: vulnerabilities, challenges and opportunities](#)

Bo Zhou, ..., and P. Lukowicz. ArXiv, 2024. 3 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) have made significant advances in natural language processing, but their underlying mechanisms are often misunderstood. Despite exhibiting coherent answers and apparent reasoning behaviors, LLMs rely on statistical patterns in word embeddings rather than true cognitive processes. This leads to vulnerabilities such as "hallucination" and misinformation. The paper argues that current LLM architectures are inherently untrustworthy due to their reliance on correlations of sequential patterns of word embedding vectors. However, ongoing research into combining generative transformer-based models with fact bases and logic programming languages may lead to the development of trustworthy LLMs capable of generating statements based on given...

[126] [Modelling Compositionality and Structure Dependence in Natural Language](#)

K. Kaushik and Andrea E. Martin. ArXiv, 2020. 0 citations.

0% Topic Match

Abstract: Human beings possess the most sophisticated computational machinery in the known universe. We can understand language of rich descriptive power, and communicate in the same environment with astonishing clarity. Two of the many contributors to the interest in natural language - the properties of Compositionality and Structure Dependence, are well documented, and offer a vast space to ask interesting modelling questions. The first step to begin answering these questions is to ground verbal theory in formal terms. Drawing on linguistics and set theory, a formalisation of these ideas is presented in the first half of this thesis. We see how...

[127] [Tense Bees and Shell-Shocked Crabs: Are Animals Conscious?](#)

M. Tye. Unknown journal, 2016. 62 citations.

0% Topic Match

No summary or abstract available

[128] [A Systematic Review on the Evaluation of Large Language Models in Theory of Mind Tasks](#)

Karahan Saritas, ..., and Yavuz Durmazkeser. ArXiv, 2025. 5 citations.

0% Topic Match

Abstract: In recent years, evaluating the Theory of Mind (ToM) capabilities of large language models (LLMs) has received significant attention within the research community. As the field rapidly evolves, navigating the diverse approaches and methodologies has become increasingly complex. This systematic review synthesizes current efforts to assess LLMs' ability to perform ToM tasks, an essential aspect of human cognition involving the attribution of mental states to oneself and others. Despite notable advancements, the proficiency of LLMs in ToM remains a contentious issue. By categorizing benchmarks and tasks through a taxonomy rooted in cognitive science, this review critically examines evaluation techniques, prompting...

[129] [Causal-aware Large Language Models: Enhancing Decision-Making Through Learning, Adapting and Acting](#)

Wei Chen, ..., and Ruichu Cai. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: Large language models (LLMs) have shown great potential in decision-making due to the vast amount of knowledge stored within the models. However, these pre-trained models are prone to lack reasoning abilities and are difficult to adapt to new environments, further hindering their application to complex real-world tasks. To address these challenges, inspired by the human cognitive process, we propose Causal-Aware LLMs, which integrate the structural causal model (SCM) into the decision-making process to model, update, and utilize structured knowledge of the environment in a "learning-adapting-acting" paradigm. Specifically, in the learning stage, we first utilize an LLM to extract the environment-specific causal entities...

[130] [Response Uncertainty and Probe Modeling: Two Sides of the Same Coin in LLM Interpretability?](#)

Yongjie Wang, ..., and Zhiqi Shen. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Probing techniques have shown promise in revealing how LLMs encode human-interpretable concepts, particularly when applied to curated datasets. However, the factors governing a dataset's suitability for effective probe training are not well-understood. This study hypothesizes that probe performance on such datasets reflects characteristics of both the LLM's generated responses and its internal feature space. Through quantitative analysis of probe performance and LLM response uncertainty across a series of tasks, we find a strong correlation: improved probe performance consistently corresponds to a reduction in response uncertainty, and vice versa. Subsequently, we delve deeper into this correlation through the lens of feature...

[131] [Babbling stochastic parrots? On reference and reference change in large language models](#)

Steffen Koch. Journal Not Provided, Unknown year. 1 citations.

0% Topic Match

Abstract: Recently developed large language models (LLMs) perform surprisingly well in many language-related tasks, ranging from text correction or authentic chat experiences to the production of entirely new texts or even essays. It is natural to get the impression that LLMs know the meaning of natural language expressions and can use them productively. Recent scholarship, however, has questioned the validity of this impression, arguing that LLMs are ultimately incapable of understanding and producing meaningful texts. This paper develops a more optimistic view. Drawing on classic externalist accounts of reference, it argues that LLMs are very likely capable of reference. Not only...

[132] [A predictive human model of language challenges traditional views in linguistics and pretrained transformer research](#)

S. Torres-Martínez. Language and Semiotic Studies, 2024. 6 citations.

0% Topic Match

Abstract: Abstract This paper introduces a theory of mind that positions language as a cognitive tool in its own right for the optimization of biological fitness. I argue that human language reconstruction of reality results from biological memory and adaptation to uncertain environmental conditions for the reaffirmation of the Self-as-symbol. I demonstrate that pretrained language models, such as ChatGPT, lack embodied grounding, which compromises their ability to adequately model the world through language due to the absence of subjecthood and conscious states for event recognition and partition. At a deep level, I challenge the notion that the constitution of a semiotic...

[133] [Towards A Holistic Landscape of Situated Theory of Mind in Large Language Models](#)

Ziqiao Ma, ..., and Joyce Chai. Unknown journal, 2023. 28 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) have generated considerable interest and debate regarding their potential emergence of Theory of Mind (ToM). Several recent inquiries reveal a lack of robust ToM in these models and pose a pressing demand to develop new benchmarks, as current ones primarily focus on different aspects of ToM and are prone to shortcuts and data leakage. In this position paper, we seek to answer two road-blocking questions: (1) How can we taxonomize a holistic landscape of machine ToM? (2) What is a more effective evaluation protocol for machine ToM? Following psychological studies, we taxonomize machine ToM into 7...

[134] [Evaluating Large Language Models on the Frame and Symbol Grounding Problems: A Zero-shot Benchmark](#)

Shoko Oka. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Recent advancements in large language models (LLMs) have revitalized philosophical debates surrounding artificial intelligence. Two of the most fundamental challenges - namely, the Frame Problem and the Symbol Grounding Problem - have historically been viewed as unsolvable within traditional symbolic AI systems. This study investigates whether modern LLMs possess the cognitive capacities required to address these problems. To do so, I designed two benchmark tasks reflecting the philosophical core of each problem, administered them under zero-shot conditions to 13 prominent LLMs (both closed and open-source), and assessed the quality of the models' outputs across five trials each. Responses were scored...

[135] [Should We Fear Large Language Models? A Structural Analysis of the Human Reasoning System for Elucidating LLM Capabilities and Risks Through the Lens of Heidegger's Philosophy](#)

Jianqiu Zhang. ArXiv, 2024. 2 citations.

0% Topic Match

Abstract: In the rapidly evolving field of Large Language Models (LLMs), there is a critical need to thoroughly analyze their capabilities and risks. Central to our investigation are two novel elements. Firstly, it is the innovative parallels between the statistical patterns of word relationships within LLMs and Martin Heidegger's concepts of "ready-to-hand" and "present-at-hand," which encapsulate the utilitarian and scientific attitudes humans employ in interacting with the world. This comparison lays the groundwork for positioning LLMs as the digital counterpart to the Faculty of Verbal Knowledge, shedding light on their capacity to emulate certain facets of human reasoning. Secondly, a structural analysis of human reasoning...

[136] [Can ChatGPT Defend its Belief in Truth? Evaluating LLM Reasoning via Debate](#)

Boshi Wang, ..., and Huan Sun. Unknown journal, 2023. 89 citations.

0% Topic Match

Abstract: Large language models (LLMs) such as ChatGPT and GPT-4 have shown impressive performance in complex reasoning tasks. However, it is difficult to know whether the models are reasoning based on deep understandings of truth and logic, or leveraging their memorized patterns in a relatively superficial way. In this work, we explore testing LLMs' reasoning by engaging with them in a debate-like conversation, where given a question, the LLM and the user need to discuss to make the correct decision starting from opposing arguments. Upon mitigating the Clever Hans effect, our task requires the LLM to not only achieve the correct...

[137] [Do Large Language Models Advocate for Inferentialism?](#)

Yuzuki Arai and Sho Tsugawa. ArXiv, 2024. 1 citations.

0% Topic Match

Abstract: The emergence of large language models (LLMs) such as ChatGPT and Claude presents new challenges for philosophy of language, particularly regarding the nature of linguistic meaning and representation. While LLMs have traditionally been understood through distributional semantics, this paper explores Robert Brandom's inferential semantics as an alternative foundational framework for understanding these systems. We examine how key features of inferential semantics -- including its anti-representationalist stance, logical expressivism, and quasi-compositional approach -- align with the architectural and functional characteristics of Transformer-based LLMs. Through analysis of the ISA (Inference, Substitution, Anaphora) approach, we demonstrate that LLMs exhibit fundamentally anti-representationalist properties in...

[138] [T RACKING THE PROGRESS OF LANGUAGE MODELS BY EXTRACTING THEIR UNDERLYING KNOWLEDGE GRAPHS](#)

No author found. Journal Not Provided, 2020. 2 citations.

0% Topic Match

Abstract: The state of the art of language models, previously dominated by pre-trained word embeddings, is now being pushed forward by large pre-trained contextual representations. This success has driven growing interest to understand what these models encode inside their inner workings. Despite this, understanding their semantic skills has been elusive, often leading to unsuccessful, non-conclusive, or contradictory results among different works. In this work, we define a probing classifier that we use to extract the underlying knowledge graph of nine of the currently most influential language models, including word embeddings, context encoders, and text generators. This probe is based on concept...

[139] [Theory of Mind in Large Language Models: Assessment and Enhancement](#)

Ruirui Chen, ..., and Cheston Tan. Unknown journal, 2025. 2 citations.

0% Topic Match

Abstract: Theory of Mind (ToM)-the ability to reason about the mental states of oneself and others-is a cornerstone of human social intelligence. As Large Language Models (LLMs) become increasingly integrated into daily life, understanding their ability to interpret and respond to human mental states is crucial for enabling effective interactions. In this paper, we review LLMs' ToM capabilities by analyzing both evaluation benchmarks and enhancement strategies. For evaluation, we focus on recently proposed and widely used story-based benchmarks. For enhancement, we provide an in-depth analysis of recent methods aimed at improving LLMs' ToM abilities. Furthermore, we outline promising directions for future research to...

[140] [The neural correlates of consciousness under the free energy principle: From computational correlates to computational explanation](#)

W. Wiese and K. Friston. Philosophy and the Mind Sciences, 2020. 25 citations.

0% Topic Match

Abstract: How can the free energy principle contribute to research on neural correlates of consciousness, and to the scientific study of consciousness more generally? Under the free energy principle, neural correlates should be defined in terms of neural "dynamics", not neural states, and should be complemented by the research on "computational" correlates of consciousness -- defined in terms of probabilities encoded by neural states. We argue that these restrictions brighten the prospects of a computational explanation of consciousness, by addressing two central problems. The first is to account for consciousness in the absence of sensory stimulation and behaviour. The second is to...

[141] [How Causal Abstraction Underpins Computational Explanation](#)

Atticus Geiger, ..., and Thomas Icard. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: Explanations of cognitive behavior often appeal to computations over representations. What does it take for a system to implement a given computation over suitable representational vehicles within that system? We argue that the language of causality -- and specifically the theory of causal abstraction -- provides a fruitful lens on this topic. Drawing on current discussions in deep learning with artificial neural networks, we illustrate how classical themes in the philosophy of computation and cognition resurface in contemporary machine learning. We offer an account of computational implementation grounded in causal abstraction, and examine the role for representation in the resulting...

[142] [Locating and Editing Factual Associations in GPT](#)

Kevin Meng, ..., and Yonatan Belinkov. Neural Information Processing Systems, 2022. 1736 citations.

Not measured Topic Match

Abstract: We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE)...

[143] [What's in the Image? A Deep-Dive into the Vision of Vision Language Models](#)

Omri Kaduri, ..., and Tali Dekel. 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 19 citations.

Not measured Topic Match

Abstract: Vision-Language Models (VLMs) have recently demonstrated remarkable capabilities in comprehending complex visual content. However, the mechanisms underlying how VLMs process visual information remain largely unexplored. In this paper, we conduct a thorough empirical analysis, focusing on the attention modules across layers, by which we reveal several key insights about how these models process visual data: (i) the internal representation of the query tokens (e.g., representations of "describe the image"), is utilized by the model to store global image information; we demonstrate that the model generates surprisingly descriptive responses solely from these tokens, without direct access to image tokens. (ii) Cross-modal...

[144] [Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking](#)

Nikhil Prakash, ..., and David Bau. ArXiv, 2024. 89 citations.

Not measured Topic Match

Abstract: Fine-tuning on generalized tasks such as instruction following, code generation, and mathematics has been shown to enhance language models' performance on a range of tasks. Nevertheless, explanations of how such fine-tuning influences the internal computations in these models remain elusive. We study how fine-tuning affects the internal mechanisms implemented in language models. As a case study, we explore the property of entity tracking, a crucial facet of language comprehension, where models fine-tuned on mathematics have substantial performance gains. We identify the mechanism that enables entity tracking and show that (i) in both the original model and its fine-tuned versions primarily...

[145] [Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small](#)

Kevin Wang, ..., and J. Steinhart. ArXiv, 2022. 705 citations.

Not measured Topic Match

Abstract: Research in mechanistic interpretability seeks to explain behaviors of machine learning models in terms of their internal components. However, most previous work either focuses on simple behaviors in small models, or describes complicated behaviors in larger models with broad strokes. In this work, we bridge this gap by presenting an explanation for how GPT-2 small performs a natural language task called indirect object identification (IOI). Our explanation encompasses 26 attention heads grouped into 7 main classes, which we discovered using a combination of interpretability approaches relying on causal interventions. To our knowledge, this investigation is the largest end-to-end attempt at...

[146] [Entity Tracking in Language Models](#)

Najoung Kim and Sebastian Schuster. Unknown journal, 2023. 26 citations.

Not measured Topic Match

Abstract: Keeping track of how states of entities change as a text or dialog unfolds is a key prerequisite to discourse understanding. Yet, there have been few systematic investigations into the ability of large language models (LLMs) to track discourse entities. In this work, we present a task probing to what extent a language model can infer the final state of an entity given an English description of the initial state and a series of state-changing operations. We use this task to first investigate whether Flan-T5, GPT-3 and GPT-3.5 can track the state of entities, and find that only GPT-3.5 models,...

[147] [Designing and Interpreting Probes with Control Tasks](#)

John Hewitt and Percy Liang. ArXiv, 2019. 574 citations.

Not measured Topic Match

Abstract: Probes, supervised models trained to predict properties (like parts-of-speech) from representations (like ELMo), have achieved high accuracy on a range of linguistic tasks. But does this mean that the representations encode linguistic structure or just that the probe has learned the linguistic task? In this paper, we propose control tasks, which associate word types with random outputs, to complement linguistic tasks. By construction, these tasks can only be learned by the probe itself. So a good probe, (one that reflects the representation), should be selective, achieving high linguistic task accuracy and low control task accuracy. The selectivity of a probe...

[148] [Inference-Time Intervention: Eliciting Truthful Answers from a Language Model](#)

Kenneth Li, ..., and M. Wattenberg. ArXiv, 2023. 732 citations.

Not measured Topic Match

Abstract: We introduce Inference-Time Intervention (ITI), a technique designed to enhance the "truthfulness" of large language models (LLMs). ITI operates by shifting model activations during inference, following a set of directions across a limited number of attention heads. This intervention significantly improves the performance of LLaMA models on the TruthfulQA benchmark. On an instruction-finetuned LLaMA called Alpaca, ITI improves its truthfulness from 32.5% to 65.1%. We identify a tradeoff between truthfulness and helpfulness and demonstrate how to balance it by tuning the intervention strength. ITI is minimally invasive and computationally inexpensive. Moreover, the technique is data efficient: while approaches like RLHF require extensive...

[149] [Understanding the Limits of Vision Language Models Through the Lens of the Binding Problem](#)

Declan Campbell, ..., and Taylor Webb. ArXiv, 2024. 39 citations.

Not measured Topic Match

Abstract: Recent work has documented striking heterogeneity in the performance of state-of-the-art vision language models (VLMs), including both multimodal language models and text-to-image models. These models are able to describe and generate a diverse array of complex, naturalistic images, yet they exhibit surprising failures on basic multi-object reasoning tasks -- such as counting, localization, and simple forms of visual analogy -- that humans perform with near perfect accuracy. To better understand this puzzling pattern of successes and failures, we turn to theoretical accounts of the binding problem in cognitive science and neuroscience, a fundamental problem that arises when a shared set...

[150] [Can Language Models Encode Perceptual Structure Without Grounding? A Case Study in Color](#)

Mostafa Abdou, ..., and Anders Søgaard. Unknown journal, 2021. 147 citations.

Not measured Topic Match

Abstract: Pretrained language models have been shown to encode relational information, such as the relations between entities or concepts in knowledge-bases — (Paris, Capital, France). However, simple relations of this type can often be recovered heuristically and the extent to which models implicitly reflect topological structure that is grounded in world, such as perceptual structure, is unknown. To explore this question, we conduct a thorough case study on color. Namely, we employ a dataset of monolexic color terms and color chips represented in CIELAB, a color space with a perceptually meaningful distance metric. Using two methods of evaluating the structural alignment...

[151] [GPT-4o System Card](#)

OpenAI Aaron Hurst, ..., and Yury Malkov. ArXiv, 2024. 2122 citations.

Not measured Topic Match

Abstract: GPT-4o is an autoregressive omni model that accepts as input any combination of text, audio, image, and video, and generates any combination of text, audio, and image outputs. It's trained end-to-end across text, vision, and audio, meaning all inputs and outputs are processed by the same neural network. GPT-4o can respond to audio inputs in as little as 232 milliseconds, with an average of 320 milliseconds, which is similar to human response time in conversation. It matches GPT-4 Turbo performance on text in English and code, with significant improvement on text in non-English languages, while also being much faster and...

[152] [Understanding intermediate layers using linear classifier probes](#)

Guillaume Alain and Yoshua Bengio. ArXiv, 2016. 1080 citations.

Not measured Topic Match

Abstract: Neural network models have a reputation for being black boxes. We propose to monitor the features at every layer of a model and measure how suitable they are for classification. We use linear classifiers, which we refer to as "probes", trained entirely independently of the model itself. This helps us better understand the roles and dynamics of the intermediate layers. We demonstrate how this can be used to develop a better intuition about models and to diagnose potential problems. We apply this technique to the popular models Inception v3 and Resnet-50. Among other things, we observe experimentally that the linear...

[153] [Probing Classifiers: Promises, Shortcomings, and Advances](#)

Yonatan Belinkov. Computational Linguistics, 2021. 534 citations.

Not measured Topic Match

Abstract: Probing classifiers have emerged as one of the prominent methodologies for interpreting and analyzing deep neural network models of natural language processing. The basic idea is simple—a classifier is trained to predict some linguistic property from a model's representations—and has been used to examine a wide variety of models and properties. However, recent studies have demonstrated various methodological limitations of this approach. This squib critically reviews the probing classifiers framework, highlighting their promises, shortcomings, and advances.

[154] [Analysis Methods in Neural Language Processing: A Survey](#)

Yonatan Belinkov and James R. Glass. Transactions of the Association for Computational Linguistics, 2018. 569 citations.

Not measured Topic Match

Abstract: The field of natural language processing has seen impressive progress in recent years, with neural network models replacing many of the traditional systems. A plethora of new models have been proposed, many of which are thought to be opaque compared to their feature-rich counterparts. This has led researchers to analyze, interpret, and evaluate neural networks in novel and more fine-grained ways. In this survey paper, we review analysis methods in neural language processing, categorize them according to prominent research trends, highlight existing limitations, and point to potential directions for future work.

[155] [Linguistic Regularities in Continuous Space Word Representations](#)

Tomas Mikolov, ..., and G. Zweig. Unknown journal, 2013. 3746 citations.

Not measured Topic Match

No summary or abstract available

[156] [Transformer Feed-Forward Layers Are Key-Value Memories](#)

Mor Geva, ..., and Omer Levy. ArXiv, 2020. 1039 citations.

Not measured Topic Match

Abstract: Feed-forward layers constitute two-thirds of a transformer model's parameters, yet their role in the network remains under-explored. We show that feed-forward layers in transformer-based language models operate as key-value memories, where each key correlates with textual patterns in the training examples, and each value induces a distribution over the output vocabulary. Our experiments show that the learned patterns are human-interpretable, and that lower layers tend to capture shallow patterns, while upper layers learn more semantic ones. The values complement the keys' input patterns by inducing output distributions that concentrate probability mass on tokens likely to appear immediately after each pattern,...

[157] [Connectionism and cognitive architecture: A critical analysis](#)

J. Fodor and Z. Pylyshyn. Cognition, 1988. 4009 citations.

Not measured Topic Match

No summary or abstract available

[158] [BERT Rediscovered the Classical NLP Pipeline](#)

Ian Tenney, ..., and Ellie Pavlick. Unknown journal, 2019. 1522 citations.

Not measured Topic Match

Abstract: Pre-trained text encoders have rapidly advanced the state of the art on many NLP tasks. We focus on one such model, BERT, and aim to quantify where linguistic information is captured within the network. We find that the model represents the steps of the traditional NLP pipeline in an interpretable and localizable way, and that the regions responsible for each step appear in the expected sequence: POS tagging, parsing, NER, semantic roles, then coreference. Qualitative analysis reveals that the model can and often does adjust this pipeline dynamically, revising lower-level decisions on the basis of disambiguating information from higher-level representations.

[159] [How Do Transformers Learn Variable Binding in Symbolic Programs?](#)

Yiwei Wu, ..., and Raphael Milliere. ArXiv, 2025. 5 citations.

Not measured Topic Match

Abstract: Variable binding -- the ability to associate variables with values -- is fundamental to symbolic computation and cognition. Although classical architectures typically implement variable binding via addressable memory, it is not well understood how modern neural networks lacking built-in binding operations may acquire this capacity. We investigate this by training a Transformer to dereference queried variables in symbolic programs where variables are assigned either numerical constants or other variables. Each program requires following chains of variable assignments up to four steps deep to find the queried value, and also contains irrelevant chains of assignments acting as distractors. Our analysis reveals...

[160] [Circuit Distillation](#)

Somin Wadhwa, ..., and Byron C. Wallace. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: Model distillation typically focuses on behavioral mimicry, where a student model is trained to replicate a teacher's output while treating its internal computations as a black box. In this work we propose an alternative approach: Distilling the underlying computational mechanisms implemented by a teacher model. Specifically, we propose circuit distillation, which introduces an objective to align internal representations between analogous circuit components in teacher and student models. We propose a method to match "functionally correspondent" circuit components and introduce a loss reflecting similarities between the representations that these induce. We evaluate circuit distillation on entity tracking and theory of mind (ToM)...

[161] [Shape Happens: Automatic Feature Manifold Discovery in LLMs via Supervised Multi-Dimensional Scaling](#)

Federico Tiblias, ..., and Iryna Gurevych. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: The linear representation hypothesis states that language models (LMs) encode concepts as directions in their latent space, forming organized, multidimensional manifolds. Prior efforts focus on discovering specific geometries for specific features, and thus lack generalization. We introduce Supervised Multi-Dimensional Scaling (SMDS), a model-agnostic method to automatically discover feature manifolds. We apply SMDS to temporal reasoning as a case study, finding that different features form various geometric structures such as circles, lines, and clusters. SMDS reveals many insights on these structures: they consistently reflect the properties of the concepts they represent; are stable across model families and sizes; actively support reasoning...

[162] [Behavioural vs. Representational Systematicity in End-to-End Models: An Opinionated Survey](#)

Ivan Vegner, ..., and Leonidas A. A. Doumas. ArXiv, 2025. 1 citations.

Not measured Topic Match

Abstract: A core aspect of compositionality, systematicity is a desirable property in ML models as it enables strong generalization to novel contexts. This has led to numerous studies proposing benchmarks to assess systematic generalization, as well as models and training regimes designed to enhance it. Many of these efforts are framed as addressing the challenge posed by Fodor and Pylyshyn. However, while they argue for systematicity of representations, existing benchmarks and models primarily focus on the systematicity of behaviour. We emphasize the crucial nature of this distinction. Furthermore, building on Hadley's (1994) taxonomy of systematic generalization, we analyze the extent to...

[163] [From superposition to sparse codes: interpretable representations in neural networks](#)

David Klindt, ..., and Nina Miolane. ArXiv, 2025. 1 citations.

Not measured Topic Match

Abstract: Understanding how information is represented in neural networks is a fundamental challenge in both neuroscience and artificial intelligence. Despite their nonlinear architectures, recent evidence suggests that neural networks encode features in superposition, meaning that input concepts are linearly overlaid within the network's representations. We present a perspective that explains this phenomenon and provides a foundation for extracting interpretable representations from neural activations. Our theoretical framework consists of three steps: (1) Identifiability theory shows that neural networks trained for classification recover latent features up to a linear transformation. (2) Sparse coding methods can extract disentangled features from these representations by leveraging...

[164] [Follow the Flow: On Information Flow Across Textual Tokens in Text-to-Image Models](#)

Guy Kaplan, ..., and Roy Schwartz. ArXiv, 2025. 2 citations.

Not measured Topic Match

Abstract: Text-to-image (T2I) models generate images by encoding text prompts into token representations, which then guide the diffusion process. While prior work has largely focused on improving alignment by refining the diffusion process, we focus on the textual encoding stage. Specifically, we investigate how semantic information is distributed across token representations within and between lexical items (i.e., words or expressions conveying a single concept) in the prompt. We analyze information flow at two levels: (1) in-item representation-whether individual tokens represent their lexical item, and (2) cross-item interaction-whether information flows across the tokens of different lexical items. We use patching techniques to...

[165] [Relational Composition in Neural Networks: A Survey and Call to Action](#)

Martin Wattenberg and Fernanda Vi'egas. ArXiv, 2024. 13 citations.

Not measured Topic Match

Abstract: Many neural nets appear to represent data as linear combinations of "feature vectors." Algorithms for discovering these vectors have seen impressive recent success. However, we argue that this success is incomplete without an understanding of relational composition: how (or whether) neural nets combine feature vectors to represent more complicated relationships. To facilitate research in this area, this paper offers a guided tour of various relational mechanisms that have been proposed, along with preliminary analysis of how such mechanisms might affect the search for interpretable features. We end with a series of promising areas for empirical research, which may help determine how neural...

[166] [LINEAR MECHANISMS FOR SPATIOTEMPORAL REASONING IN VISION LANGUAGE MODELS](#)

Raphi Kang, ..., and Pietro Perona. Journal Not Provided, Unknown year. 0 citations.

Not measured Topic Match

No summary or abstract available

[167] [Mechanistic Interpretability for AI Safety - A Review](#)

Leonard Bereska and E. Gavves. ArXiv, 2024. 201 citations.

Not measured Topic Match

Abstract: Understanding AI systems' inner workings is critical for ensuring value alignment and safety. This review explores mechanistic interpretability: reverse engineering the computational mechanisms and representations learned by neural networks into human-understandable algorithms and concepts to provide a granular, causal understanding. We establish foundational concepts such as features encoding knowledge within neural activations and hypotheses about their representation and computation. We survey methodologies for causally dissecting model behaviors and assess the relevance of mechanistic interpretability to AI safety. We examine benefits in understanding, control, alignment, and risks such as capability gains and dual-use concerns. We investigate challenges surrounding scalability, automation, and...

[168] [RAVEL: Evaluating Interpretability Methods on Disentangling Language Model Representations](#)

Jing Huang, ..., and Atticus Geiger. ArXiv, 2024. 50 citations.

Not measured Topic Match

Abstract: Individual neurons participate in the representation of multiple high-level concepts. To what extent can different interpretability methods successfully disentangle these roles? To help address this question, we introduce RAVEL (Resolving Attribute-Value Entanglements in Language Models), a dataset that enables tightly controlled, quantitative comparisons between a variety of existing interpretability methods. We use the resulting conceptual framework to define the new method of Multi-task Distributed Alignment Search (MDAS), which allows us to find distributed representations satisfying multiple causal criteria. With Llama2-7B as the target language model, MDAS achieves state-of-the-art results on RAVEL, demonstrating the importance of going beyond neuron-level analyses to...

[169] [MIB: A Mechanistic Interpretability Benchmark](#)

Aaron Mueller, ..., and Yonatan Belinkov. ArXiv, 2025. 5 citations.

Not measured Topic Match

Abstract: How can we know whether new mechanistic interpretability methods achieve real improvements? In pursuit of lasting evaluation standards, we propose MIB, a Mechanistic Interpretability Benchmark, with two tracks spanning four tasks and five models. MIB favors methods that precisely and concisely recover relevant causal pathways or causal variables in neural language models. The circuit localization track compares methods that locate the model components - and connections between them - most important for performing a task (e.g., attribution patching or information flow routes). The causal variable localization track compares methods that featurize a hidden vector, e.g., sparse autoencoders (SAEs) or distributed...

[170] [Mechanistic Interpretability of Emotion Inference in Large Language Models](#)

Ala Nekouvaht Tak, ..., and Jonathan Gratch. Unknown journal, 2025. 2 citations.

Not measured Topic Match

Abstract: Large language models (LLMs) show promising capabilities in predicting human emotions from text. However, the mechanisms through which these models process emotional stimuli remain largely unexplored. Our study addresses this gap by investigating how autoregressive LLMs infer emotions, showing that emotion representations are functionally localized to specific regions in the model. Our evaluation includes diverse model families and sizes and is supported by robustness checks. We then show that the identified representations are psychologically plausible by drawing on cognitive appraisal theory, a well-established psychological framework positing that emotions emerge from evaluations (appraisals) of environmental stimuli. By causally intervening on construed...

[171] [Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability](#)

Atticus Geiger, ..., and Thomas F. Icard. ArXiv, 2023. 87 citations.

Not measured Topic Match

Abstract: Causal abstraction provides a theoretical foundation for mechanistic interpretability, the field concerned with providing intelligible algorithms that are faithful simplifications of the known, but opaque low-level details of black box AI models. Our contributions are (1) generalizing the theory of causal abstraction from mechanism replacement (i.e., hard and soft interventions) to arbitrary mechanism transformation (i.e., functionals from old mechanisms to new mechanisms), (2) providing a flexible, yet precise formalization for the core concepts of polysemantic neurons, the linear representation hypothesis, modular features, and graded faithfulness, and (3) unifying a variety of mechanistic interpretability methods in the common language of causal...

[172] [From Emergence to Control: Probing and Modulating Self-Reflection in Language Models](#)

Xudong Zhu, ..., and Zhihui Zhu. ArXiv, 2025. 2 citations.

Not measured Topic Match

Abstract: Self-reflection -- the ability of a large language model (LLM) to revisit, evaluate, and revise its own reasoning -- has recently emerged as a powerful behavior enabled by reinforcement learning with verifiable rewards (RLVR). While self-reflection correlates with improved reasoning accuracy, its origin and underlying mechanisms remain poorly understood. In this work, {it we first show that self-reflection is not exclusive to RLVR fine-tuned models: it already emerges, albeit rarely, in pretrained models}. To probe this latent ability, we introduce Reflection-Inducing Probing, a method that injects reflection-triggering reasoning traces from fine-tuned models into pretrained models. This intervention raises self-reflection frequency...

[173] [Mechanistic evaluation of Transformers and state space models](#)

Aryaman Arora, ..., and Christopher Potts. ArXiv, 2025. 3 citations.

Not measured Topic Match

Abstract: State space models (SSMs) for language modelling promise an efficient and performant alternative to quadratic-attention Transformers, yet show variable performance on recalling basic information from the context. While performance on synthetic tasks like Associative Recall (AR) can point to this deficiency, behavioural metrics provide little information as to why--on a mechanistic level--certain architectures fail and others succeed. To address this, we conduct experiments on AR and find that only Transformers and Based SSM models fully succeed at AR, with Mamba a close third, whereas the other SSMs (H3, Hyena) fail. We then use causal interventions to explain why. We find...

[174] [Illusion or Algorithm? Investigating Memorization, Emergence, and Symbolic Processing in In-Context Learning](#)

Jingcheng Niu, ..., and Iryna Gurevych. ArXiv, 2025. 1 citations.

Not measured Topic Match

Abstract: Large-scale Transformer language models (LMs) trained solely on next-token prediction with web-scale data can solve a wide range of tasks after seeing just a few examples. The mechanism behind this capability, known as in-context learning (ICL), remains both controversial and poorly understood. Some studies argue that it is merely the result of memorizing vast amounts of data, while others contend that it reflects a fundamental, symbolic algorithmic development in LMs. In this work, we introduce a suite of investigative tasks and a novel method to systematically investigate ICL by leveraging the full Pythia scaling suite, including interim checkpoints that capture...

[175] [Dense SAE Latents Are Features, Not Bugs](#)

Xiaoqing Sun, ..., and Max Tegmark. ArXiv, 2025. 4 citations.

Not measured Topic Match

Abstract: Sparse autoencoders (SAEs) are designed to extract interpretable features from language models by enforcing a sparsity constraint. Ideally, training an SAE would yield latents that are both sparse and semantically meaningful. However, many SAE latents activate frequently (i.e., are \emph{dense}), raising concerns

that they may be undesirable artifacts of the training procedure. In this work, we systematically investigate the geometry, function, and origin of dense latents and show that they are not only persistent but often reflect meaningful model representations. We first demonstrate that dense latents tend to form antipodal pairs that reconstruct specific directions in the residual stream, and...

[176] [Evaluating Large Language Models for Real-World Engineering Tasks](#)

René Heesch, ..., and Oliver Niggemann. ArXiv, 2025. 1 citations.

Not measured Topic Match

Abstract: Large Language Models (LLMs) are transformative not only for daily activities but also for engineering tasks. However, current evaluations of LLMs in engineering exhibit two critical shortcomings: (i) the reliance on simplified use cases, often adapted from examination materials where correctness is easily verifiable, and (ii) the use of ad hoc scenarios that insufficiently capture critical engineering competencies. Consequently, the assessment of LLMs on complex, real-world engineering problems remains largely unexplored. This paper addresses this gap by introducing a curated database comprising over 100 questions derived from authentic, production-oriented engineering scenarios, systematically designed to cover core competencies such as product...

[177] [Large Language Models Share Representations of Latent Grammatical Concepts Across Typologically Diverse Languages](#)

Jannik Brinkmann, ..., and Aaron Mueller. ArXiv, 2025. 20 citations.

Not measured Topic Match

Abstract: Human bilinguals often use similar brain regions to process multiple languages, depending on when they learned their second language and their proficiency. In large language models (LLMs), how are multiple languages learned and encoded? In this work, we explore the extent to which LLMs share representations of morphosyntactic concepts such as grammatical number, gender, and tense across languages. We train sparse autoencoders on Llama-3-8B and Aya-23-8B, and demonstrate that abstract grammatical concepts are often encoded in feature directions shared across many languages. We use causal interventions to verify the multilingual nature of these representations; specifically, we show that ablating only...

[178] [Does Object Binding Naturally Emerge in Large Pretrained Vision Transformers?](#)

Yihao Li, ..., and Konrad P. Kording. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: Object binding, the brain's ability to bind the many features that collectively represent an object into a coherent whole, is central to human cognition. It groups low-level perceptual features into high-level object representations, stores those objects efficiently and compositionally in memory, and supports human reasoning about individual object instances. While prior work often imposes object-centric attention (e.g., Slot Attention) explicitly to probe these benefits, it remains unclear whether this ability naturally emerges in pre-trained Vision Transformers (ViTs). Intuitively, they could: recognizing which patches belong to the same object should be useful for downstream prediction and thus guide attention. Motivated by...

[179] [ToMMeR -- Efficient Entity Mention Detection from Large Language Models](#)

Victor Morand, ..., and Benjamin Piwowarski. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: Identifying which text spans refer to entities -- mention detection -- is both foundational for information extraction and a known performance bottleneck. We introduce ToMMeR, a lightweight model (<300K parameters) probing mention detection capabilities from early LLM layers. Across 13 NER benchmarks, ToMMeR achieves 93% recall zero-shot, with over 90% precision using an LLM as a judge showing that ToMMeR rarely produces spurious predictions despite high recall. Cross-model analysis reveals that diverse architectures (14M-15B parameters) converge on similar mention boundaries (DICE>75%), confirming that mention detection emerges naturally from language modeling. When extended with span classification heads, ToMMeR achieves near SOTA...

[180] [Achilles' Heel of Mamba: Essential difficulties of the Mamba architecture demonstrated by synthetic data](#)

Tianyi Chen, ..., and Zhi Xu. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: State Space Models (SSMs) have emerged as promising alternatives to attention mechanisms, with the Mamba architecture demonstrating impressive performance and linear complexity for processing long sequences. However, the fundamental differences between Mamba and Transformer architectures remain incompletely understood. In this work, we use carefully designed synthetic tasks to reveal Mamba's inherent limitations. Through experiments, we identify that Mamba's nonlinear convolution introduces an asymmetry bias that significantly impairs its ability to recognize symmetrical patterns and relationships. Using composite function and inverse sequence matching tasks, we demonstrate that Mamba strongly favors compositional solutions over symmetrical ones and struggles with tasks requiring the...

[181] [Position: We Need An Algorithmic Understanding of Generative AI](#)

Oliver Eberle, ..., and Ida Momennejad. ArXiv, 2025. 4 citations.

Not measured Topic Match

Abstract: What algorithms do LLMs actually learn and use to solve problems? Studies addressing this question are sparse, as research priorities are focused on improving performance through scale, leaving a theoretical and empirical gap in understanding emergent algorithms. This position paper proposes AlgEval: a framework for systematic research into the algorithms that LLMs learn and use. AlgEval aims to uncover algorithmic primitives, reflected in latent representations, attention, and inference-time compute, and their algorithmic composition to solve task-specific problems. We highlight potential methodological paths and a case study toward this goal, focusing on emergent search algorithms. Our case study illustrates both the...

[182] [Understanding In-context Learning of Addition via Activation Subspaces](#)

Xinyan Hu, ..., and Lijie Chen. ArXiv, 2025. 5 citations.

Not measured Topic Match

Abstract: To perform in-context learning, language models must extract signals from individual few-shot examples, aggregate these into a learned prediction rule, and then apply this rule to new examples. How is this implemented in the forward pass of modern transformer models? To study this, we consider a structured family of few-shot learning tasks for which the true prediction rule is to add an integer k to the input. We find that Llama-3-8B attains high accuracy on this task for a range of k , and localize its few-shot ability to just three attention heads via a novel optimization approach. We further show...